HPSC FSO Master Handout: Food Processing Operations & Commodity Technology

Read First: Method & Corrections

HPSC Food Safety Officer — Master Study Handout

Module: Food Processing Operations & Commodity Technology (Dairy, Cereals, Meat/Fish/Poultry/Egg, Horticulture)

1. How to use this handout

This handout teaches each commodity from first principles in a fixed order: physical-chemical nature of the raw material → engineering unit operation → machinery parameters → biochemical transformation → diagnostic quality defect → statutory or trade limit → descriptive answer skeleton. Every section is written out in full. Numerical values are given with their source basis (physics, textbook convention, FSSR/BIS/USDA standard) so that you can defend them in a descriptive answer and avoid being trapped by an altered number in an objective question.

Three labels recur throughout:

- **Principle:** a physical, chemical or biological law that will not change between editions of a textbook.
- **Convention:** a widely taught number or range that varies slightly between references (for example 8–10 percent bran versus 5–8 percent bran in paddy). Learn the ordering and the magnitude; quote the figure used by the examiner's likely source.
- **Standard:** a regulatory or grading figure (FSSR, BIS, USDA, EU). Verify against the currently notified text before quoting it as law.

2. Examination architecture (working assumptions)

Stage	Supplied pattern	Consequence for preparation
Stage 1 Objective Screening	100 marks; 0.25 negative marking; count-based pair items; absolute-qualifier traps	Learn exact numbers, units, directions (increase/decrease) and organism-process pairs
Stage 2 Subject Knowledge Test	150 marks; 3 hours; conventional descriptive answers; stated 87.5 percent merit weightage	Master mechanisms, flow sheets, comparative tables and labelled diagrams described in words

These particulars are user-supplied working assumptions. The notification applicable to the actual recruitment cycle controls.

3. Negative-marking arithmetic

With one mark for a correct response and 0.25 deducted for a wrong one, the expected score of an attempt with probability p of being correct is

$$E = p - 0.25(1 - p) = 1.25p - 0.25$$

Break-even is $p = 0.20$. In a four-option item, eliminating a single option already lifts p to 0.33 and makes the attempt positive in expectation. Eliminate on units, direction and organism-process logic before guessing.

4. Count-based pair protocol

For every three-pair item, score each pair independently as $c_i = 1$ (correct) or $c_i = 0$ (incorrect), then compute

$$N = c_1 + c_2 + c_3$$

Map N to the option: 0 → None; 1 → Only 1; 2 → Only 2; 3 → All 3. The option counts correct pairs; it never names which pair. Never let the correctness of pair 1 colour your judgement of pair 2.

5. Descriptive-answer skeleton (use for every 10- and 15-mark answer)

1. Definition and first-principle intuition (one short paragraph).
2. Classification or flow sheet (numbered sequence or comparison table).
3. Mechanism: physics, machinery parameter, or biochemical pathway.
4. Numbers with units and their basis.
5. Defects, diagnostics and control measures.
6. One-sentence conclusion linking back to food safety or quality.

Time budget: 150 marks in 180 minutes gives 1.2 minutes per mark; reserve 15 minutes for reading and review, leaving roughly 11 minutes for a 10-mark answer and 16–17 minutes for a 15-mark answer.

6. Correction and qualification register (read before memorising any number)

Brief statement in the syllabus outline	Safe examination interpretation
Milk freezing point −0.530 to −0.550 °C	Typical range; many references quote −0.512 to −0.550 °C. Adulteration with water moves the value towards 0 °C (for example −0.48 °C).
Paneer coagulation at 80–85 °C	Milk is first heated to about 90 °C, then coagulated at 70–85 °C with 1–2 percent citric acid at pH 5.3–5.4. The pH figure matters more than the exact coagulation temperature.
PSE: pH < 5.4 while warm	Standard diagnostic is a rapid fall to pH ≤ 5.8–6.0 within 45 minutes post-mortem while muscle is still above about 35 °C; ultimate pH may then reach 5.3–5.5. The combination of low pH and high temperature is the defining feature, not any single pH number.
DFD: ultimate pH > 6.2	Many references use > 6.0; 6.2 is a commonly taught threshold. Learn the direction: DFD = high ultimate pH.
Candling Grade B air cell > 8 mm	USDA uses depth greater than 3/16 inch (about 4.8 mm) for Grade B; some Indian texts quote 8 mm. The ordering AA (≤ 3 mm) < A < B is what is tested.
Paddy bran 8–10 percent	Conventions range 5–10 percent; husk 20–22 percent; endosperm about 70 percent; germ 2–3 percent.
Khoa compositional standard	FSSR requires milk fat not less than 30 percent on dry-matter basis; moisture varies by type (Pindi lowest, Dhap highest). Verify the exact total-solids figure from the currently notified FSSR before quoting.
Nitrite 100–150 ppm	This is the usual ingoing level; statutory maxima (residual or ingoing) differ between jurisdictions. Quote ingoing level and purpose, then the applicable legal limit if asked.

When an objective item quotes a number that differs slightly from the one you memorised, test the direction and the magnitude first. A pair is usually made wrong by reversing a relationship (break versus reduction roll speed ratio), swapping an organism (Proteus versus Pseudomonas rot), or changing a unit (minutes versus seconds), far more often than by shifting a number by a few percent.

7. Reading order

Part 1 (Dairy) → Part 2 (Cereals and Pulses) → Part 3 (Meat, Fish and Egg) → Part 4 (Horticulture) → Revision Bank and Model Answers. Each part ends with its own HPSC EXAM TRAP boxes and comparative tables.

Part 1: Milk & Dairy Processing

Part 1. Milk and Dairy Processing Technology

1.1 Milk as a polyphasic system

Principle: Milk is not a single solution. It is three physical phases coexisting in one liquid, and every dairy unit operation acts on one of these phases.

Phase	Dispersed component	Particle size	What processing does to it
Emulsion (oil-in-water)	Fat globules, each surrounded by a native milk-fat-globule membrane (MFGM) of phospholipid and protein	0.1–15 µm, mean 3–4 µm	Creaming, centrifugal separation, homogenisation, churning (phase inversion)
Colloidal suspension	Casein micelles (αs1, αs2, β, κ-casein with colloidal calcium phosphate)	50–300 nm, mean about 150 nm	Rennet and acid coagulation, heat stability, UF retention
True solution	Lactose, whey proteins (β-lactoglobulin, α-lactalbumin, immunoglobulins), soluble minerals, vitamins, urea	Molecular	Freezing-point depression, osmotic pressure, NF/RO concentration, Maillard browning

Approximate composition of cow milk: water 87 percent, fat 3.5–4.5 percent, protein 3.2–3.5 percent (casein about 80 percent of protein), lactose 4.6–4.9 percent, ash 0.7 percent. Buffalo milk carries 6–8 percent fat and larger fat globules, which is why it gives higher khoa and ghee yields.

Physical constants used as adulteration indicators

Property	Normal value	Basis	Adulteration signal
Specific gravity (15.5 °C)	1.028–1.032 (lactometer reading 28–32)	Solids-not-fat raise density; fat lowers it	Added water lowers it; skimming raises it; both together can mask each other (hence combine with fat test)
Titratable acidity	0.12–0.16 percent as lactic acid (fresh milk, natural acidity from casein, phosphates, citrates, CO2)	Titration with N/10 NaOH to phenolphthalein end-point	Developed acidity above about 0.18 percent indicates bacterial souring; neutralisers (soda) lower it artificially (Rosalic acid test)
Freezing point	−0.512 to −0.550 °C, typical −0.53 to −0.55 °C	Colligative property of dissolved lactose and salts; most constant physical property of milk	Added water moves freezing point towards 0 °C. Hortvet or thermistor cryoscopy quantifies extraneous water: percent added water ≈ (T_ref − T_sample)/T_ref × 100, with T_ref taken as −0.540 °C
pH	6.6–6.8	Buffering by phosphates, citrates and proteins	Falls with souring; rises with mastitis or added alkali
Boiling point	100.17 °C	Slight elevation by solutes	Rarely used diagnostically

[HPSC EXAM TRAP] Freezing point depression, not specific gravity, is the reference method for added water, because sugar or salt added by an adulterator can restore the lactometer reading but every added solute changes the freezing point in a predictable way. A pair stating that cryoscopy detects skimming is wrong; cryoscopy detects water.

1.2 Centrifugal separation and Stokes' law

Principle: Fat globules are less dense (about 0.92 g/mL) than skim milk (about 1.036 g/mL), so they rise. The rate of rise of a single globule under gravity is given by Stokes' law:

$$v = \frac{d^2 (\rho_s - \rho_f) g}{18 \eta}$$

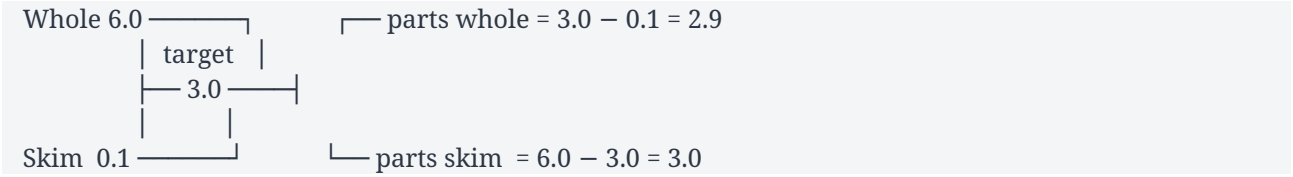
where d is globule diameter, ρ_s is serum density, ρ_f is fat density, g is gravitational acceleration and η is viscosity. Doubling the diameter quadruples the creaming rate; cooling raises viscosity and slows creaming; in raw cold milk the cryoglobulin (IgM)-mediated clustering of globules accelerates creaming far beyond the single-globule prediction.

In a centrifugal separator g is replaced by $\omega^2 r$ (angular velocity squared times radius), giving separation forces of 5,000–10,000 g at 6,000–8,000 rpm. The disc-bowl separator stacks conical discs 0.4–0.6 mm apart so that each globule travels only a short distance before reaching a disc surface. Warm milk (40–55 °C) is separated because fat is fully liquid and viscosity is low.

Outputs: cream (35–40 percent fat for butter, 20–25 percent for table cream) and skim milk (0.05–0.1 percent fat). Separation efficiency is judged by skim-milk fat; above 0.1 percent indicates worn discs, excessive throughput or cold milk. A clarifier is the same machine without a cream outlet; a bactofuge removes bacteria and spores by density difference at higher g.

Standardisation by the Pearson square

Worked example: Prepare 1,000 kg of 3.0 percent fat milk from whole milk at 6.0 percent and skim milk at 0.1 percent.



Total parts = 2.9 + 3.0 = 5.9. Whole milk = 1,000 × 2.9/5.9 = 491.5 kg; skim milk = 508.5 kg. Check: (491.5 × 6.0 + 508.5 × 0.1)/1,000 = (2,949 + 50.9)/1,000 = 3.0 percent. The rule: each component's parts equal the difference between the target and the other component's value, taken across the diagonal.

1.3 Homogenisation engineering

Principle: Creaming cannot be stopped by chemistry alone; it is stopped by making the fat globules so small that Stokes' velocity becomes negligible. Reducing mean diameter from 3–4 µm to below 1 µm (practically all below 2 µm) cuts the creaming rate by more than a factor of ten while increasing the total fat surface area 6–10 fold.

Parameter	Value	Reason
First-stage pressure	15–25 MPa (2,000–3,000 psi; 150–250 bar)	Supplies the energy for turbulence, cavitation and shear in the valve gap
Second-stage pressure	3–5 MPa (500 psi)	Breaks up the clusters of newly formed small globules that stick together through shared casein coating; essential for cream and high-fat products
Temperature	55–80 °C	Fat must be liquid; cold

Parameter	Value	Reason
		homogenisation gives poor disruption and lipolysis
Valve gap	about 0.1 mm	Velocity in the gap exceeds 100 m/s, pressure drops abruptly, cavitation bubbles collapse

New globule surfaces are coated predominantly by casein micelle fragments and some whey protein because the native MFGM is insufficient to cover a tenfold larger area.

Consequences of homogenisation

Effect	Direction	Explanation
Creaming	Prevented	Small diameter; also casein-coated globules do not form cryoglobulin clusters
Whiteness and viscosity	Increased	More light-scattering surfaces; higher dispersed-phase volume with protein coating
Curd tension	Reduced (softer curd)	Casein is partly bound to fat surfaces and the fat globules act as structure breakers in the gel; useful for infant and clinical milks
Susceptibility to lipolysis	Increased	Native MFGM is replaced by casein, which lipoprotein lipase can penetrate easily; raw homogenised milk turns rancid in minutes, so homogenisation is always paired with pasteurisation (upstream or immediately after)
Light-induced oxidised flavour	Increased	Larger surface exposes lipids; sunlight flavour
Heat stability of concentrated milk	Decreased	Casein-coated fat behaves like large casein particles
Foaming	Reduced in whole milk	Surface-active MFGM phospholipids are redistributed

[HPSC EXAM TRAP] Homogenisation does not kill bacteria and does not reduce lipase activity; it increases lipolysis risk. A pair stating that homogenisation inactivates lipase is wrong. The second stage is not for further size reduction; it is for cluster dispersion.

1.4 Thermal processing regimes

Principle: Pasteurisation is a public-health treatment designed to destroy the most heat-resistant non-sporing pathogen historically relevant to milk (Coxiella burnetii, with Mycobacterium tuberculosis as the earlier benchmark) while causing minimal chemical change. Sterilisation (UHT or in-container) destroys spores to achieve commercial sterility. Each regime has a built-in biochemical marker that is inactivated at approximately the same time-temperature combination as the target organism.

Regime	Time-temperature	Equipment	Target and validation	Shelf life
LTLT (holder/batch)	63 °C for 30 min	Jacketed vat with agitator	Pathogen destruction; alkaline phosphatase	7–10 days at ≤ 4 °C

Regime	Time-temperature	Equipment	Target and validation	Shelf life
			(ALP) negative	
HTST (continuous)	72 °C for 15 s (India: 71.7 °C for 15 s)	Plate heat exchanger with regeneration (≥ 90 percent), holding tube, flow diversion valve, booster pump	ALP negative; lactoperoxidase (Storch) remains positive, proving no over-heating	10–14 days at ≤ 4 °C
Flash / higher-heat shorter-time	89 °C for 1 s; 94 °C for 0.1 s; 100 °C for 0.01 s	Plate or tubular	ALP negative	As HTST
UHT	135–150 °C for 1–4 s (commonly 138–142 °C for 2–4 s)	Direct (steam injection or infusion, then flash cooling) or indirect (plate/tubular); aseptic filling	Commercial sterility; turbidity test negative; ALP negative but may reactivate; spores of Geobacillus stearothermophilus and Bacillus sporothermodurans are the resistance benchmarks	6 months ambient, unopened
In-container sterilisation	110–120 °C for 10–40 min in retort	Batch or continuous (hydrostatic) steriliser, glass or plastic bottles	Commercial sterility; turbidity test negative; marked browning, cooked flavour	6–12 months ambient

Validation chemistry

- **Alkaline phosphatase (ALP) test:** ALP is a native milk enzyme slightly more heat resistant than Coxiella burnetii. Its presence (phenol released from disodium phenyl phosphate, detected with 2,6-dichloroquinone chloroimide giving blue, or by the Fluorophos method) means under-pasteurisation or raw-milk contamination. Negative = properly pasteurised. Microbial ALP from psychrotrophs and ALP reactivation in UHT milk after storage are known false positives.
- **Turbidity test (Aschaffenburg):** Ammonium sulphate (20 g) is added to 20 mL milk; after filtration the filtrate is boiled. In raw or pasteurised milk, soluble whey proteins remain in the filtrate and precipitate on boiling, giving turbidity. In sterilised and UHT milk the whey proteins have already been denatured and are removed with the casein on filtration, so the boiled filtrate stays clear. Negative turbidity = adequately sterilised.
- **Lactoperoxidase (Storch) test:** Lactoperoxidase survives HTST but is destroyed above about 80 °C; a negative Storch test on milk sold as pasteurised indicates over-heating.
- **Methylene blue reduction and standard plate count:** Confirm microbial quality rather than heat treatment.

[HPSC EXAM TRAP] ALP negative proves pasteurisation, not sterilisation; turbidity negative proves sterilisation. A pair assigning the turbidity test to HTST milk is wrong. Also, UHT milk does not require refrigeration before opening; pasteurised milk does.

1.5 Membrane separation

Principle: Pressure-driven membranes fractionate milk by molecular size without heat, preserving native structure. Pore size falls and operating pressure rises as the process moves from MF to RO.

Process	Pore size / cut-off	Pressure	Retained (retentate)	Passed (permeate)	Dairy application
Microfiltration (MF)	0.1–10 µm	0.1–0.3 MPa (1–3 bar)	Bacteria, spores, somatic cells, fat globules	Casein micelles, whey proteins, lactose, salts	Extended-shelf-life (ESL) milk; bacterial and spore removal from skim milk before cheese making; casein-whey fractionation with 0.1 µm membranes
Ultrafiltration (UF)	1–100 nm; 1–500 kDa	0.2–1.0 MPa (2–10 bar)	Proteins (casein and whey) and fat	Lactose, minerals, water, non-protein nitrogen	Whey protein concentrate (WPC 35–80), protein-standardised cheese milk, pre-concentration for soft cheese (MMV process), high-protein yoghurt
Nanofiltration (NF)	about 1 nm; 200–1,000 Da	1–4 MPa (10–40 bar)	Lactose, divalent ions, proteins	Water and monovalent ions (Na^+, K^+, Cl^-)	Partial demineralisation of whey (30–40 percent), whey concentration, lactose standardisation
Reverse osmosis (RO)	< 1 nm; dense membrane	3–6 MPa (30–60 bar)	All solutes	Water only	Pre-concentration of milk or whey before evaporation (energy saving), cheese-milk concentration, water recovery

Membrane fouling (protein deposition, calcium phosphate scaling) is controlled by cross-flow velocity, temperature (about 50 °C for whey), pH and cleaning-in-place cycles. Diafiltration (adding water to the retentate and re-filtering) is used in UF to reach higher protein purity.

1.6 Concentration and spray drying

Evaporation: Milk is first concentrated to 45–55 percent total solids in multiple-effect falling-film evaporators under vacuum (boiling at 45–70 °C) because removing water by evaporation costs about one-tenth of removing it in a spray dryer. Mechanical or thermal vapour recompression lowers steam use further.

Spray-drying sequence

1. Concentrate is pumped to the atomiser at the top of a tall cylindrical chamber.
2. Atomisation converts the liquid into droplets of 10–200 µm, creating enormous surface area.
3. Hot air (inlet 180–220 °C) contacts the droplets; evaporation is so fast that the droplet surface stays near the wet-bulb temperature (about 50 °C) during the constant-rate period, protecting proteins.
4. Drying is complete in 5–30 seconds; air leaves at 75–95 °C (outlet temperature governs final moisture).

5. Powder is separated from the air in cyclones (centrifugal) with bag filters for fines; fines are returned to the atomiser zone for agglomeration.
6. A vibrating fluid bed completes drying from about 6 to 3–4 percent moisture, cools the powder and allows lecithination (0.2 percent soya lecithin) for instant whole-milk powder.

Atomiser	Mechanism	Parameters	Characteristics
Rotary (centrifugal) wheel	Liquid fed to the centre of a disc rotating at 10,000–20,000 rpm; flung off the rim as a sheet that breaks into droplets	Peripheral speed 100–200 m/s; low feed pressure	Handles viscous and abrasive feeds, wide chamber needed (radial spray), flexible capacity, more occluded air, lower bulk density
Pressure nozzle	Feed forced at 5–30 MPa through a small orifice with swirl chamber	High-pressure pump; several nozzles per dryer	Narrow spray cone, tall-form dryers, denser powder, less occluded air, susceptible to orifice wear and blockage; preferred for agglomerated instant powders
Two-fluid (pneumatic) nozzle	Compressed air shears the liquid	Low throughput	Pilot scale

Droplet drying kinetics

A droplet dries in two periods: a constant-rate period, where free water evaporates from a saturated surface and the rate depends on air temperature and humidity, and a falling-rate period, where a solid crust forms and internal diffusion of water through the lactose-protein matrix controls the rate. Rapid crust formation at excessive inlet temperature traps vapour, causing hollow or burst particles and scorched particles (brown specks measured on the ADMI scorched-particle disc).

Sticky-point temperature and lactose glass transition

Spray-dried lactose is amorphous (a glass). Above the glass-transition temperature (Tg) the matrix softens; about 10–20 °C above Tg the particles become sticky and adhere to walls and to each other. Tg of dry amorphous lactose is about 100 °C but falls steeply as moisture rises (water is a plasticiser), so a powder at 5 percent moisture can become sticky at 40–50 °C. Outlet temperature and powder moisture are therefore set below the sticky point; whey powders, rich in lactose, are pre-crystallised (seeded and held at 30 °C) to convert lactose to the non-hygroscopic α-monohydrate before drying.

Powder quality indices

Index	Method	Typical standard
Solubility (insolubility) index	10 g powder reconstituted, centrifuged; sediment in mL	Spray-dried SMP not more than 2.0 mL (FSSR; ADMI good powder ≤ 1.25 mL); roller-dried not more than 15 mL
Moisture	Oven at 102 °C	SMP and WMP not more than 5 percent (FSSR); typical 3–4 percent
Fat	Rose-Gottlieb	SMP not more than 1.5 percent; WMP not less than 26 percent
Whey protein nitrogen index (WPNI)	Undenatured whey protein, mg N per g powder	Low-heat powder ≥ 6.0; medium 1.5–6.0; high-heat ≤ 1.5 (bakery and chocolate require high-heat; cheese and recombined milk require low-heat)
Bulk density	Tapped volume	0.5–0.7 g/mL; rotary atomiser gives

Index	Method	Typical standard
		lower density
Scorched particles	ADMI disc A–D	Grade A or B acceptable
Wettability and dispersibility	Time to sink and dispersed percentage	Instant WMP wettable in ≤ 20 s

[HPSC EXAM TRAP] The solubility index is a sediment volume: a lower number means a better powder. A pair stating that high solubility index indicates good solubility is wrong. Roller-dried powder has lower solubility (higher index) because the film is heat-denatured on the drum at 130–150 °C.

1.7 Butter technology

Principle (phase inversion): Cream is an oil-in-water emulsion. Churning subjects it to air incorporation and mechanical shock; fat globules are driven to the air bubble surfaces, their membranes are stripped, partially crystalline fat spreads over the bubbles, and when the foam collapses the bare globules coalesce into butter grains. The serum (buttermilk) is expelled and the continuous phase becomes fat: a water-in-oil emulsion with 16 percent water dispersed as droplets of 1–10 µm.

Process sequence

1. Cream separation to 35–40 percent fat; neutralisation if acidity is high (to protect fat from hydrolysis during pasteurisation).
2. Pasteurisation of cream at 85–95 °C (higher than milk because fat protects microbes and lipase must be destroyed); vacreation removes feed flavours.
3. Ripening (optional, cultured butter) with Lactococcus lactis subsp. cremoris and Leuconostoc (diacetyl producers) to 0.4–0.5 percent acidity.
4. Ageing at 4–7 °C for 12–16 hours to crystallise fat; the ratio of solid to liquid fat (about 50 percent solid at churning) governs grain formation; the Alnarp tempering program (cold-warm-cold) adjusts spreadability.
5. Churning at 8–14 °C (lower in summer, higher in winter) in a batch churn at 20–30 rpm for 30–45 minutes, or in a continuous Fritz-Eisenreich machine with a high-speed beater (1,000–1,400 rpm) followed by working sections. Break point occurs when grains the size of wheat appear.
6. Draining buttermilk; washing grains with chilled water to remove curd and serum.
7. Salting (salt not more than 3 percent in table butter) and working: kneading disperses moisture into fine droplets (most below 10 µm) so that microbial growth is confined to isolated droplets, removes free water, and gives a uniform close texture. Over-working causes greasy, leaky butter; under-working causes mottled, free-moisture butter.
8. Packaging and cold storage at −20 to −25 °C for long storage.

Standards (FSSR, table butter)

Constituent	Limit
Milk fat	Not less than 80 percent
Moisture	Not more than 16 percent
Curd (milk solids-not-fat)	Not more than 1.5 percent
Common salt	Not more than 3.0 percent
Desi (cooking) butter	Milk fat not less than 76 percent

Theories tested: Fisher and Hooker (foam theory), Rahn (foam-protein denaturation), King (phase-inversion theory, now accepted). Overrun is the butter yield above the weight of fat; theoretical overrun = (butter weight − fat weight)/fat weight × 100, typically 20–24 percent.

1.8 Ghee technology

Principle: Ghee is clarified milk fat; the aim is to remove water (to below 0.5 percent) and curd so that hydrolytic and microbial spoilage stop, while heat develops the characteristic cooked flavour.

Methods of preparation

Method	Steps	Yield and remarks
Desi (indigenous)	Milk → dahi (curd) → churned to makkhan → heated to ghee	Low fat recovery (80–85 percent), high flavour, 80 percent of Indian production; lactic flavour notes
Creamery butter method	Cream → butter → melted and clarified	88–92 percent recovery; standard dairy method
Direct cream method	Cream heated and clarified directly	Saves butter-making step; slightly more cooked and less typical flavour; higher acidity
Pre-stratification	Butter melted and held at 80–85 °C for 15–30 minutes; stratifies into top scum, middle fat, bottom serum; serum (80 percent of moisture and curd) drained before final clarification	Saves fuel, lower FFA, lower phospholipid (hence slightly lower keeping quality)
Continuous	Scraped-surface heat exchangers and centrifugal clarifiers	Industrial scale

Clarification chemistry

Heating to 110–120 °C (held briefly) drives off water as the mixture froths, denatures and browns the curd (Maillard reaction with lactose) and produces the cooked, nutty ghee flavour. Flavour compounds: δ-lactones (C10–C14, from hydroxy fatty acids) give the sweet buttery note; methyl ketones (C7–C15, from β-keto acids by decarboxylation) give the characteristic ghee aroma; free fatty acids (butyric) give sharpness; carbonyls and pyrazines from Maillard reactions add the toasted note. Clarification at 140 °C gives stronger flavour and colour but lowers vitamin A and raises FFA. Phospholipids transferred from the curd residue act as natural antioxidants; this is why desi ghee keeps longer.

Analytical purity standards (FSSR)

Parameter	Limit	What it detects
Butyro-refractometer (BR) reading at 40 °C	40.0–43.0	Vegetable fats and animal body fats raise the reading (vanaspati 48–52, lard higher); extreme value signals gross adulteration
Reichert-Meissl (RM) value	Not less than 28 (24 or 21 in specified regions such as the cotton tract)	Milligrams of N/10 KOH for water-soluble volatile fatty acids (butyric, caproic) in 5 g fat; milk fat is uniquely rich (24–34); vegetable oils and body fats have RM below 1, so adulteration lowers it
Polenske value	1.0–2.0	Water-insoluble volatile fatty acids (caprylic, capric, lauric); coconut and palm-kernel oil raise it to 15–18

Parameter	Limit	What it detects
Kirschner value	Typically 20–26 in genuine ghee	Water-soluble volatile acids as their silver salts; specific for butyric acid; distinguishes milk fat from coconut oil (which raises RM slightly but gives very low Kirschner)
Moisture	Not more than 0.5 percent	Hydrolytic rancidity risk
Free fatty acids	Not more than 3.0 percent as oleic acid	Poor raw material or storage
Baudouin test	Negative	Sesame oil (mandatory 5 percent in vanaspati) gives crimson colour with furfural and HCl, proving vanaspati adulteration
Phytosterol acetate test	Negative (melting point of sterol acetate 114–115 °C for cholesterol)	Vegetable fats introduce phytosterol acetate (126–137 °C)

[HPSC EXAM TRAP] Adulteration with vegetable oil lowers the RM value and raises the BR reading; a pair stating that vanaspati raises RM is wrong. Coconut oil raises the Polenske value but not the Kirschner value. Saponification value of ghee is high (220–230) because of short-chain acids; iodine value is low (26–38).

1.9 Traditional Indian dairy products

Principle: Indian products are made by three routes: heat-acid coagulation (paneer, chhana), thermal desiccation (khoa) and fermentation (dahi). The route determines the protein structure, moisture and end use.

Product	Route and conditions	Key parameters	Standard (FSSR)	Yield and uses
Paneer	Milk (buffalo preferred, 5–6 percent fat) heated to 90 °C, cooled to 70–85 °C, coagulated with 1–2 percent citric acid (or sour whey) to pH 5.3–5.4; curd settled, whey drained, pressed at 2–3 kg/cm² for 15–20 minutes, chilled in water at 4–6 °C	Casein and denatured whey proteins co-precipitate (hence higher yield than cheese); pressing gives a firm sliceable block that holds shape on frying; too low a pH gives hard, crumbly paneer; too high gives soft, high-moisture paneer	Moisture not more than 70 percent; milk fat not less than 50 percent on dry-matter basis (FDM); low-fat paneer: fat not more than 15 percent FDM	20–22 percent from buffalo milk; 16–18 percent from cow milk; curries, snacks
Chhana	Cow milk preferred (smaller fat globules, softer casein, less calcium) heated to boiling, cooled to 70–80 °C, coagulated with lactic or citric acid at pH 5.4; whey drained through muslin without pressing	Soft, moist, granular, spongy body with good kneadability; buffalo milk gives hard, greasy chhana unsuitable for rasogolla	Moisture not more than 70 percent; fat not less than 50 percent FDM	16–18 percent; rasogolla, sandesh, chhana-murki, rasmalai (Bengali sweets)
Khoa	Milk boiled vigorously in a shallow open karahi with constant scraping; moisture evaporates, proteins denature, lactose concentrates and	Buffalo milk preferred (higher fat, softer body, whiter colour, less sandiness)	Milk fat not less than 30 percent on dry-matter basis; moisture generally 20–30 percent depending on type	Buffalo milk 20–23 percent; cow milk 17–19 percent; burfi, peda, gulab jamun, kalakand

Product	Route and conditions	Key parameters	Standard (FSSR)	Yield and uses
	partly crystallises, fat partially separates; pat formed at about 70 percent total solids; also made in continuous scraped-surface heat exchangers and inclined steam-jacketed pans			
Dahi	Standardised milk heated to 85–90 °C, cooled to 30–37 °C, inoculated 1–2 percent with mixed mesophilic lactococci and thermophilic lactobacilli, incubated 8–12 hours	Acid gel of casein at pH 4.4–4.6; diacetyl flavour	Acidity, set body; probiotic variants	Mishti doi (sweetened), lassi, shrikhand (chakka)

Khoa types

Type	Moisture	Texture	Sweet made
Pindi	Lowest (about 20–25 percent)	Hard, compact, circular pat	Burfi, peda
Dhap	Highest (35–40 percent)	Loose, sticky, high moisture	Gulab jamun, pantua
Danedar	Intermediate	Granular; a little acid (sour whey or citric acid) is added during heating to induce controlled coagulation	Kalakand, kunda

Khoa defects: sandiness (lactose crystals after over-concentration or slow cooling), browning (Maillard, high temperature), rancidity and mould on storage (water activity 0.9 permits growth; shelf life 2–3 days ambient, 2 weeks refrigerated).

1.10 Comparative table: fluid-milk heat processes at a glance

Feature	LTLT	HTST	UHT
Temperature and time	63 °C, 30 min	72 °C, 15 s	135–150 °C, 1–4 s
Microbial target	Vegetative pathogens	Vegetative pathogens	Spores; commercial sterility
Marker enzyme	ALP negative	ALP negative, lactoperoxidase positive	Turbidity negative; ALP negative (may reactivate)
Whey protein denaturation	5–10 percent	10–15 percent	70–90 percent (indirect)
Flavour	Slight	Minimal	Cooked, sulphurous (direct UHT milder)
Packaging and storage	Refrigerated	Refrigerated	Aseptic, ambient
Typical plant	Vat	Plate heat exchanger	Steam injection or tubular with aseptic filler

Part 2: Cereals, Pulses & Grain Technology

Part 2. Cereal, Pulse and Grain Technology

2.1 Rice (Oryza sativa) processing

Anatomy of paddy

Principle: Milling is the mechanical separation of the layers of the grain. Every yield figure in rice milling follows from the proportion of each layer.

Layer	Share of paddy weight (convention)	Composition and function	Removed at
Husk (hull: lemma and palea)	20–22 percent	Silica-rich, inedible, abrasive; used as fuel and for furfural	Dehusking (shelling)
Bran (pericarp, seed coat, nucellus, aleurone)	5–10 percent (commonly quoted 8–10)	Fat 15–20 percent, protein, fibre, B-vitamins, minerals; aleurone carries lipase	Whitening and polishing
Germ (embryo)	2–3 percent	Fat and protein rich	Comes off with bran
Endosperm	About 70 percent	Starch (90 percent of endosperm dry matter; amylose 15–30 percent), protein 7–8 percent (glutelin oryzenin)	This is the milled rice

Theoretical yields: brown rice 78–80 percent of paddy; milled (white) rice 68–72 percent of paddy. Raw paddy milling in India often yields only 62–65 percent, mainly because of breakage.

Milling flow sheet

1. **Cleaning:** sieves, aspirators and destoners remove straw, stones, dust, immature grains.
2. **Dehusking (shelling):** Rubber roll sheller — two rubber-covered rolls rotating in opposite directions at different speeds (differential about 25 percent; for example 1,100 and 850 rpm) with a gap of about 0.5–0.8 mm; the shear between rolls splits the husk off with minimal breakage; 85–90 percent shelling efficiency per pass. Older Engelberg (huller) and disc (emery) shellers cause heavy breakage and bran loss and are discouraged. Husk is aspirated off; the mixture of brown rice and unshelled paddy goes to a paddy separator (tray or compartment type, using differences in density, size and surface friction) and unshelled paddy is returned.
3. **Whitening (bran removal):** (a) abrasive whiteners (Satake vertical cone): a rotating emery cone inside a perforated screen grinds the bran off; low pressure, suitable for hard parboiled rice and the first pass; (b) friction whiteners (horizontal): grains rub against each other and against a ribbed steel rotor at higher pressure; the bran peels off; gives a smoother surface, better for raw rice and the final passes. Modern mills use multi-pass abrasive followed by friction, with bran aspiration to keep the grain cool.
4. **Polishing:** a soft leather or cloth-lined cone or water-mist polisher removes adhering bran dust and gives gloss (mist polishing). Over-polishing lowers yield and nutrients.
5. **Grading:** indented cylinders or plan-sifters separate head rice, large brokens, small brokens (brewers) and bran.

Key milling indices

- **Head rice yield (HRY):** weight of whole kernels (length ≥ three-quarters of the intact kernel) as a percentage of paddy; the single most important economic index because brokens sell at half price. Raw rice HRY 50–60 percent; parboiled rice 65–70 percent.
- **Degree of milling (DOM) / degree of polishing:** percentage of bran removed relative to brown rice weight; 8–10 percent is full milling; under-milled 4–6 percent. Measured by weight loss, surface lipid content, or whiteness meter. Higher DOM lowers thiamine (about 80 percent of thiamine is in the bran and germ) and fat.
- **Milling recovery:** total milled rice as a percentage of paddy.
- Breakage is caused by fissures (sun-cracks from rapid field drying, moisture re-adsorption), chalky kernels, immature grain and high milling pressure; optimum paddy moisture for milling is 13–14 percent.

Parboiling: hydrothermal processing

Principle: Soaking, steaming and drying paddy before milling gelatinises the starch inside the husk. On drying, the starch re-solidifies as a hard, translucent, fissure-free mass, so the grain resists breakage and the bran is held more tightly, while soluble nutrients migrate inward.

Step	Conditions (CFTRI hot-soaking method)	Purpose
Cleaning and soaking	Hot water 65–70 °C for 3–4 hours (raises paddy moisture to about 30 percent); traditional cold soaking takes 2–3 days and causes fermentation odour	Hydrate the endosperm uniformly
Steaming	Saturated steam at 100 °C (atmospheric) for 5–10 minutes, or pressure parboiling at 1–2 kg/cm²	Gelatinise starch (gelatinisation temperature 60–75 °C), kill fungi and insects, inactivate enzymes
Drying	Sun or mechanical drying to 12–14 percent moisture, ideally in two stages with tempering to avoid stress cracks	Re-solidify starch, allow safe storage and milling
Milling	As raw rice but with higher whitening pressure (harder grain)	Bran comes off in larger flakes, preferred for bran-oil extraction

Biochemical changes: starch gelatinisation and partial retrogradation on drying; amylose-lipid complex formation (lowers glycemic index, reduces cooking loss and stickiness); migration of thiamine, niacin and minerals from the aleurone into the endosperm (thiamine retention 70–80 percent in parboiled milled rice against 20–30 percent in raw milled rice; this is why parboiling protected rice-eating populations from beriberi); protein denaturation and surface protein film, which reduces solids loss in cooking water; inactivation of lipase (preventing hydrolytic rancidity of the bran, so parboiled bran keeps better and gives better oil); sealing of fissures and hardening (HRY rises to 65–70 percent); husk loosening; colour change to amber-yellow from Maillard reactions and diffusion of husk pigments; development of a characteristic parboiled flavour; insect resistance from hardness and lower free sugars.

Advantages	Disadvantages
Higher head rice yield (less breakage)	Colour darkening and off-flavour if soaking is prolonged
Higher thiamine and mineral retention	Longer cooking time (more energy)
Lipase inactivation; better bran oil	Extra capital, fuel and water; effluent
Lower cooking loss, non-sticky grains, higher swelling	Bran is harder to remove; consumes more whitening energy
Better storage and insect resistance	Vitamin losses from heat if over-processed
Easier husk removal	Mould growth and fermentation in traditional cold soaking

Advantages	Disadvantages

[HPSC EXAM TRAP] Parboiling is done on paddy (with husk), never on brown or milled rice; the husk is what allows the vitamins to be trapped inside. A pair stating that parboiling removes bran is wrong: bran removal is milling. Parboiling increases, not decreases, the B-vitamin content of milled rice.

2.2 Wheat milling

Wheat kernel

Bran (pericarp, testa, aleurone) about 14–15 percent; endosperm about 83 percent (starch 70 percent, protein 10–12 percent as gluten-forming gliadin and glutenin); germ 2–3 percent (fat 10 percent, enzymes, vitamin E). Hard wheat (Triticum aestivum hard varieties) has strong adhesion between starch and protein and fractures along cell walls to give gritty flour and strong gluten for bread; soft wheat gives fine, weak-gluten flour for biscuits and cakes.

Cleaning

Magnetic separator (iron), milling separator (sieves and aspiration for straw, dust), destoner (gravity table), disc or cylinder separator (seeds of other crops, by length), scourer (removes beeswing and surface dirt), washer-whizzer (optional). Cleaning is called the screen room.

Conditioning and tempering

Principle: The miller wants the bran to come off as large, tough, unbroken flakes and the endosperm to crumble easily. Water does both: it toughens the bran (plasticises the fibrous pericarp so it does not shatter into fine specks that contaminate the flour) and mellows the endosperm (weakens the protein-starch bond).

Wheat type	Target moisture	Resting (tempering) time
Hard wheat	15.5–16.5 percent	18–24 hours (water must diffuse to the centre)
Soft wheat	14.0–15.0 percent	6–12 hours
Durum for semolina	16.0–17.0 percent	12–24 hours (very hard, long diffusion)

Warm conditioning (water at 45 °C) shortens tempering; hot conditioning above 60 °C damages gluten. Over-wetting makes the mill choke and the flour mouldy; under-wetting gives bran powder in the flour and high ash.

Roller milling: the gradual reduction system

Principle: The mill does not grind wheat in one stroke. It opens the grain, scrapes endosperm from the bran in stages (break system), classifies the stock by size and purity (sifting and purifying), and only then grinds the clean endosperm particles (semolina, middlings) into flour (reduction system). Each pair of rolls is followed by a sifter.

Feature	Break rolls	Reduction rolls
Surface	Corrugated (fluted), with flutes becoming finer from the first break (about 10–12 flutes per inch) to the fifth break (about 26–28 per inch);	Smooth (frosted or matt finish)

Feature	Break rolls	Reduction rolls
	flutes set sharp-to-sharp or dull-to-dull	
Speed differential (fast roll : slow roll)	About 2.5 : 1 (fast roll 500–600 rpm)	About 1.25 : 1 (lower differential)
Mechanical action	Shearing and cutting: the fast fluted roll shears the kernel open against the slow roll, scraping endosperm chunks away from the bran flake	Crushing and compression: particles are squeezed between smooth surfaces to flour size with minimal bran fragmentation
Number of passages	4–6 breaks (B1 to B5 or B6)	8–12 reductions (C1 to C12), plus sizings and tailings rolls
Feed	B1 receives whole tempered wheat; later breaks receive the overtails (bran with adhering endosperm) from the previous sifter	Purified semolina and middlings (sizings)
Products	Coarse bran flakes, semolina (coarse endosperm), middlings (fine endosperm), a little break flour	Flour (patent and clear grades), fine bran and shorts tailings, germ (flattened and sifted out because it is oily and plastic)
Roll gap	Opens progressively; B1 about 0.5 mm, B5 much tighter	Very fine, set so that starch damage is controlled (6–9 percent damaged starch is desirable for bread flour water absorption)

Sifters (plansifters): stacked sieves of decreasing aperture (wire and nylon cloth) that separate each roll's output into overtails (to the next break), semolina and middlings (to purifiers), and throughs (flour).

Purifiers: an inclined oscillating sieve deck with a controlled upward current of air. Semolina from the break system still carries bran chips of the same size. Air aspiration lifts the lighter bran particles while the denser endosperm passes through the graded sieves; the purified semolina goes to the first reduction rolls. The purifier is the device that makes high-grade white flour (low ash) possible.

Flour streams and extraction rate

Extraction rate = weight of flour obtained per 100 parts of cleaned wheat.

Product	Extraction (convention)	Description and use
Whole wheat atta (chakki or roller)	95–100 percent	All bran and germ retained; ash 1.5–2.0 percent; chapati and roti; chakki (stone) milling damages more starch and gives better chapati texture
Resultant atta	About 90 percent	Wholemeal recombined after removal of some bran
Maida (refined, straight-grade white flour)	70–72 percent	Ash below 0.6–0.7 percent (FSSR ash not more than 1.0 percent on dry basis, gluten not less than 7.5 percent, moisture not more than 14 percent — verify current text); bread, biscuits, cakes
Patent flour	40–60 percent (the purest central-endosperm streams)	Ash 0.35–0.45 percent; premium bread flour
Clear flour	Remainder of white flour after patent	Higher ash, darker; rye-type breads, gluten extraction
Semolina (suji / rava)	Coarse endosperm particles 250–500 µm, recovered before reduction	Upma, halwa, pasta (from durum)
Bran	14–15 percent	Feed, fibre supplement

Product	Extraction (convention)	Description and use
Shorts and germ	2–4 percent	Feed, wheat-germ oil

Flour treatments: ageing (natural oxidation of thiol groups improves gluten), bleaching and maturing agents (benzoyl peroxide, ascorbic acid as dough improver; chlorine for cake flour; potassium bromate withdrawn in India), enrichment (iron, folic acid, vitamin B12 for fortified atta and maida), α-amylase supplementation (malt flour or fungal amylase) for gassing power.

Durum wheat (Triticum durum)

Attribute	Durum	Bread wheat (T. aestivum)
Ploidy	Tetraploid, 2n = 28 (AABB genome)	Hexaploid, 2n = 42 (AABBDD)
Protein	12–15 percent, high	10–13 percent
Gluten character	Strong but less elastic, more extensible; the D-genome glutenins that give bread elasticity are absent, so durum makes poor bread but excellent pasta	Elastic, gas-retaining
Endosperm	Very hard, vitreous (translucent); vitreousness above 80 percent is a grade criterion and predicts high semolina yield	Mealy to vitreous
Pigment	High lutein (xanthophyll carotenoid), giving amber-yellow semolina and pasta; low lipoxygenase activity is selected to protect lutein during pasta making	Low
Milling objective	Maximise coarse semolina (65–70 percent extraction), minimise flour; more break passages, coarser flutes, extensive purification, high tempering moisture	Maximise flour
Pasta extrusion	Semolina + water to 30–32 percent moisture, mixed under vacuum (prevents air bubbles and lutein oxidation), extruded at 8–12 MPa and 40–50 °C through bronze or Teflon dies (spaghetti, macaroni), dried in stages at 50–90 °C to 12.5 percent moisture; protein network and starch form a compact glassy matrix; high-temperature drying strengthens cooking quality and reduces stickiness	Not applicable

[HPSC EXAM TRAP] Break rolls are corrugated with the higher speed differential (2.5:1); reduction rolls are smooth with the lower differential (1.25:1). Items swap one of these four attributes. Also, lutein is a carotenoid pigment, not a protein; vitreousness is a physical endosperm property, not a protein content.

2.3 Maize (corn) wet milling

Principle: Wet milling is not milling for flour; it is a fractionation of the kernel into its chemically distinct components (starch, protein, germ oil, fibre) using water, mild chemistry and density differences. Dry milling, by contrast, produces grits, meal and flour for food.

Maize kernel: endosperm 82 percent (starch 70 percent of kernel, protein zein and glutelin), germ 11–12 percent (fat 33 percent; source of corn oil), pericarp (bran) 5–6 percent, tip cap 1 percent.

Flow sheet

1. **Cleaning:** remove cob pieces, dust, stones, other seeds.
2. **Steeping:** kernels soaked in counter-current steep tanks at 48–52 °C for 30–48 hours in water containing 0.1–0.2 percent sulphur dioxide (as sulphurous acid) and lactic acid (produced by Lactobacillus during steeping, about 1–2 percent). Functions of SO2: (a) cleaves disulphide bonds of the protein matrix (zein and glutelin) surrounding starch granules, loosening the granules; (b) suppresses putrefactive and spore-forming bacteria while permitting lactic fermentation; (c) prevents enzymatic browning; (d) softens the kernel to 45 percent moisture. Steepwater is drawn off, concentrated to 50 percent solids as **corn steep liquor (CSL)** — rich in soluble protein, lactic acid, B-vitamins and minerals; used as a fermentation nutrient (penicillin, citric acid production) and in gluten feed.
3. **First grinding (degermination):** coarse attrition (Bauer) mills tear the softened kernel open without breaking the oil-rich germ.
4. **Germ separation:** in hydrocyclones (formerly flotation tanks), the oil-rich germ (specific gravity below 1.0) rises in the starch slurry (specific gravity about 1.08) and is removed overhead. Germ is washed, dried and pressed or solvent-extracted for **corn oil** (crude, then refined; high in linoleic acid and tocopherols); the germ cake goes to feed.
5. **Fine grinding and fibre screening:** the degermed slurry is ground finely in impact or attrition mills; fibre (bran) is screened off on fine bend screens (DSM) and washed counter-currently; fibre plus CSL becomes corn gluten feed (21 percent protein).
6. **Starch-gluten separation:** the mill starch (starch + gluten protein) is sent to continuous centrifuges (primary separators). Gluten protein (specific gravity about 1.1) is lighter than starch (1.5) and comes off as the overflow; it is thickened, dewatered and dried to **corn gluten meal** (60 percent protein, rich in xanthophyll; poultry feed pigment).
7. **Starch washing:** the starch underflow is washed in 8–14 stages of hydrocyclones to remove residual protein (below 0.3 percent) and solubles.
8. **Starch finishing:** dewatered on vacuum filters and flash-dried to 12 percent moisture (native starch), or modified (acid-thinned, cross-linked, oxidised, pre-gelatinised), or converted to sweeteners.

Yield per 100 kg maize (dry basis, approximate): starch 66–68 kg, gluten feed 20 kg, gluten meal 5–6 kg, oil 3–4 kg.

High-fructose corn syrup (HFCS) manufacture

Step	Enzyme and conditions	Product
Liquefaction	Thermostable bacterial α-amylase (Bacillus licheniformis), jet cooking at 105–110 °C, pH 6.0–6.5, calcium added; starch slurry 30–35 percent solids	Dextrins, dextrose equivalent (DE) 10–15
Saccharification	Fungal glucoamylase (Aspergillus niger) with pullulanase (debranching), 60 °C, pH 4.0–4.5, 24–72 hours	Glucose syrup, DE 95–98
Refining	Filtration, activated carbon, ion exchange	Clear, colourless dextrose syrup
Isomerisation	Immobilised glucose isomerase (Streptomyces), 55–60 °C, pH 7.5–8.0, Mg2+ activator, continuous column	HFCS-42 (42 percent fructose, 53 percent glucose) — equilibrium conversion
Enrichment	Chromatographic (simulated moving bed) separation on calcium-form resin	HFCS-90, blended with HFCS-42 to HFCS-55 (sweetness equal to sucrose;

Step	Enzyme and conditions	Product
		soft drinks)

Dextrose equivalent (DE) measures reducing power relative to pure glucose (100); maltodextrin is below 20 DE, glucose syrup above 20 DE. Other starch derivatives: dextrose monohydrate (crystallised), maltodextrin, modified starches, sorbitol (by hydrogenation), dextrins (by dry roasting).

[HPSC EXAM TRAP] The germ is separated by density (it floats) in hydrocyclones; starch and gluten are separated by density in centrifuges (gluten is lighter and leaves as overflow). A pair stating that starch is lighter than gluten is wrong. Steeping uses sulphur dioxide, not sodium hydroxide; the alkaline process (nixtamalisation with lime) is a different, food-use treatment of maize for tortillas that improves niacin availability.

2.4 Pulse processing and milling

Principle: A pulse grain is two cotyledons (85–90 percent of grain) wrapped in a seed coat (husk, 10–15 percent) that is attached by a layer of gums and mucilage, strongest in pigeon pea (arhar) and weakest in chickpea and lentil. Dal milling must loosen this gummy bond before the husk can be removed without crushing the cotyledons. Loosening is done by water, oil and heat; removal is done by abrasion; splitting of the cotyledons follows naturally once the husk is gone.

Why dehull?

Removes the fibrous, tannin-rich seed coat (improves digestibility, cooking time, appearance and reduces polyphenol antinutrients), and splitting shortens cooking. Dehulling does not remove trypsin inhibitors or oligosaccharides (raffinose, stachyose, verbascose: flatulence factors) — these need soaking, cooking or germination.

Methods

Feature	Dry (oil) milling — traditional and CFTRI	Wet milling (water conditioning)
Pre-treatment	Cleaning and grading; pitting (scratching the seed coat by passing once through an emery roller) so that oil can penetrate	Cleaning; soaking in water for 2–12 hours (grain absorbs 8–10 percent moisture)
Conditioning agent	Edible oil (linseed or groundnut) applied at 0.1–0.3 percent and mixed; heaped and left overnight (tempering); then sun-dried 2–4 days (or CFTRI hot-air conditioning at 120 °C for 10–15 minutes) to about 10 percent moisture; oil and heat hydrolyse and dehydrate the gum layer and create a gap between husk and cotyledon	Water swells the cotyledons more than the husk, loosening the bond; grain is then sun-dried, a small amount of oil (0.1–0.2 percent) may be applied, and dried further
Dehusking	Emery-coated or carborundum roller (under-runner disc sheller or Chakki); husk aspirated; several passes (gota = dehusked whole, dal = split)	Same roller or disc; fewer passes because the husk is already loose
Cotyledon splitting	Passing the dehusked gota through a gentle roller or by impact; the two cotyledons separate at the natural plane	Same
Dal yield	70–75 percent (theoretical maximum about 85–88 percent); CFTRI improved	72–78 percent; better for hard-to-mill pulses (pigeon pea, black gram)

Feature	Dry (oil) milling — traditional and CFTRI	Wet milling (water conditioning)
	method 75–80 percent	
Milling loss	10–15 percent as powder (chuni), brokens and over-abraded cotyledon; higher in traditional multi-pass milling	Lower powder loss, but higher energy and time; risk of fermentation if soaked too long; colour and cooking quality may suffer
Polishing	Water, oil or leather polishing for shine (Tur dal is often polished with water and soapstone powder; avoid non-permitted colours)	Same

Parameters and defects

- **Dehulling efficiency** = (whole dehulled grains + splits without husk)/total × 100.
- **Broken and powder** should be below 5–8 percent; excessive moisture or pressure increases brokens.
- **Theoretical yield** = 100 − husk fraction; the gap between theoretical and actual yield is the milling loss and is the main economic index of a dal mill.
- **Mini dal mills** (CFTRI, PKV) for village level use single emery rollers with 1–2 percent oil and 8–10 percent moisture conditioning.
- **Antinutritional factors** removed or reduced by processing: trypsin inhibitors and lectins (heat labile, destroyed by boiling or autoclaving at 120 °C for 15–30 minutes), tannins and phytates (reduced by dehulling, soaking and germination), oligosaccharides (reduced by soaking, germination, fermentation), lathyrogen β-ODAP in khesari dal (Lathyrus sativus; reduced by soaking in hot water and discarding water; sale regulated under FSSR).

[HPSC EXAM TRAP] Oil in dry pulse milling is a conditioning aid at 0.1–0.3 percent, not a frying or flavouring step; and heating is for loosening the husk, not for destroying trypsin inhibitors (which requires moist heat during cooking). Dehulling lowers tannins and fibre but does not remove flatulence oligosaccharides, which are in the cotyledon.

2.5 Comparative table: wheat fractions and extraction

Fraction	Approximate share of grain	Ash	Protein	Principal use
Patent flour	40–60	0.35–0.45	11–12 (hard wheat)	Premium bread, cake (soft)
Straight-grade maida	70–72	0.5–0.7	10–12	General bakery
Semolina / suji	Recovered at 10–20 percent of break stock	0.6–0.8	11–13 (durum higher)	Pasta, upma
Whole atta	95–100	1.5–2.0	11–13	Chapati
Bran	14–15	5–7	14–16	Feed, fibre
Germ	2–3	4–5	25–30	Oil, supplements

Part 3: Meat, Poultry, Fish & Egg

Part 3. Meat, Poultry, Fish and Egg Processing

3.1 Muscle biology

Principle: Meat is skeletal muscle whose living machinery (contraction, energy supply, oxygen transport) has stopped. Everything that happens after slaughter is the running down of that machinery without a blood supply; tenderness, colour and water-holding capacity are the visible results.

Structural hierarchy

Muscle → bundles (fasciculi, wrapped in perimysium) → fibres (cells, wrapped in endomysium, 10–100 µm diameter) → myofibrils (1–2 µm) → sarcomeres (repeating units between Z-lines, 2–3 µm in resting muscle) → thick filaments (myosin) and thin filaments (actin with tropomyosin and troponin). The whole muscle is sheathed in epimysium (connective tissue) continuous with tendon.

Muscle proteins

Class	Share of muscle protein	Members	Solubility	Significance in meat
Myofibrillar (contractile)	55–60 percent	Myosin (about 45 percent of myofibrillar; ATPase head), actin (20 percent), tropomyosin, troponin, titin, nebulin, actinins	Salt-soluble (extracted at 0.3–0.6 M NaCl; the basis of phosphate-salt brines and emulsion sausage binding)	Rigor mortis (actomyosin), water-holding capacity, gel formation in sausages, texture
Sarcoplasmic	30–35 percent	Myoglobin (colour), haemoglobin residue, glycolytic enzymes (glyceraldehyde-3-phosphate dehydrogenase, aldolase, creatine kinase), lysosomal cathepsins, calpains	Water-soluble (low ionic strength)	Colour, post-mortem glycolysis, drip loss (denatured sarcoplasmic proteins in PSE), flavour precursors
Stromal (connective tissue)	10–15 percent	Collagen (types I and III; triple helix, hydroxyproline marker, converts to gelatin on moist heating at 60–70 °C), elastin (heat-stable, yellow, in ligaments), reticulin	Insoluble	Background toughness; increases with animal age because of heat-stable cross-links (pyridinoline); hydroxyproline assay estimates collagen content in meat products

Myoglobin colour chemistry: deoxymyoglobin (purple-red, Fe2+), oxymyoglobin (bright cherry red, Fe2+ with O2; the bloom seen on freshly cut meat), metmyoglobin (brown, Fe3+; oxidation at low oxygen tension, favoured by high pH drop, light and bacteria). Cooked meat: denatured globin haemichrome (brown-grey). Cured meat: nitrosomyoglobin (red) → nitrosohaemochrome (pink) on cooking.

3.2 Post-mortem physiology

Glycolysis without oxygen

After exsanguination, oxygen delivery stops. The muscle continues to need ATP to maintain ion gradients and to keep the actin-myosin cross-bridges relaxed. Creatine phosphate regenerates ATP for a few minutes; thereafter the only route is anaerobic glycolysis of stored glycogen (0.5–1.3 percent of muscle):

$$\text{Glycogen} \rightarrow \text{Glucose-1-phosphate} \rightarrow \ldots \rightarrow \text{Pyruvate} \rightarrow \text{Lactic acid} + \text{ATP}$$

Because there is no circulation, lactic acid accumulates and pH falls from about 7.0–7.2 in living muscle to an ultimate pH (pHu) of 5.4–5.8 in normal beef, pork and lamb within 18–24 hours (beef) or 6–8 hours (pork), and 2–4 hours in poultry breast. Glycolysis stops when glycogen is exhausted or when the pH inactivates phosphofructokinase, whichever comes first. The lower the pH, the closer proteins are to their isoelectric point (about 5.1–5.3 for myofibrillar proteins), the lower the net charge and the lower the water-holding capacity (WHC). The pH decline is also what makes fresh meat microbiologically more stable than live tissue and what gives cooked meat its typical flavour profile.

Rigor mortis mechanics

Phase	Biochemical state	Physical state	Timing (beef / pork / poultry)
Delay (pre-rigor)	ATP maintained above about 1 µmol/g by creatine phosphate and glycolysis; cross-bridges cycle freely	Muscle soft, extensible; pre-rigor meat has highest WHC and best salt-extractability (hot boning advantage)	0–6 h / 0–1 h / 0–0.5 h
Onset	ATP falls below about 1 µmol/g; calcium leaks from sarcoplasmic reticulum (pump fails); myosin heads bind actin but cannot detach without ATP	Progressive stiffening, loss of extensibility	6–12 h / 1–3 h / 0.5–1 h
Full rigor	ATP exhausted; permanent actomyosin cross-bridges (rigor complex); sarcomeres may shorten	Maximum stiffness and toughness; minimum WHC	12–24 h / 3–6 h / 1–4 h
Resolution (conditioning, ageing)	Proteolysis of the cytoskeletal scaffold (titin, nebulin, desmin, troponin-T) by calpains (µ- and m-calpain, calcium-activated neutral proteases, active at pH 6–7 early post-mortem; regulated by calpastatin) and cathepsins (lysosomal acid proteases B, D, H, L released as pH falls); the actin-myosin bonds themselves are not reversed, but the structure around them is weakened	Gradual tenderisation; WHC partially recovers; flavour develops	7–21 days at 1–4 °C (beef) / 3–5 days / 1 day

Rigor shortening is worst when ATP exhaustion occurs at very high or very low temperatures: **cold shortening** (muscle chilled below 10 °C while pH is still above 6.0 and ATP is present; calcium pump fails in the cold, massive contraction up to 40 percent, severe toughness; beef and lamb) and **thaw rigor** (pre-rigor frozen muscle contracts violently on thawing). Both are prevented by electrical stimulation of the

carcass (accelerates glycolysis so rigor is complete before chilling), conditioning at 15 °C until pH falls below 6, and pelvic (tenderstretch) suspension.

[HPSC EXAM TRAP] Resolution of rigor is not the breaking of actomyosin cross-bridges by ATP (ATP never returns); it is enzymatic weakening of the surrounding structure by calpains and cathepsins. Rigor mortis in fish and poultry begins and resolves much faster than in beef. Items that assign calpains to acid conditions and cathepsins to neutral conditions have reversed them.

3.3 Muscle defects: PSE and DFD

Principle: Both defects are abnormalities in the rate or extent of post-mortem glycolysis, and both stem from pre-slaughter stress; the difference is whether the stress was acute (minutes to an hour before slaughter) or chronic (hours to days).

Feature	PSE (pale, soft, exudative)	DFD (dark, firm, dry; dark-cutting beef)	Normal
Species	Mainly pork; also turkey and chicken breast	Mainly beef, lamb; also pork	—
Stress type	Acute: excitement, fighting, rough handling or electrical goading immediately before slaughter; genetic predisposition (halothane-sensitive RYR1 ryanodine-receptor mutation, porcine stress syndrome)	Chronic: long transport, fasting, cold, fear, mixing of strange animals, exhaustion over 12–48 hours	Rested
Muscle glycogen at death	Normal or high	Depleted before slaughter	Normal
Rate of pH fall	Very rapid: pH below 5.8–6.0 within 45 minutes while carcass is still warm (above 35 °C)	Very slow or absent (little glycogen to convert)	Gradual over 6–24 h
Ultimate pH	Normal to low (5.3–5.6)	High: above 6.0 (6.2 commonly quoted)	5.4–5.8
Mechanism of defect	Low pH at high temperature denatures sarcoplasmic and myofibrillar proteins; denatured proteins precipitate onto myofibrils, scatter light (pale) and lose water-binding (exudative, soft)	High pH keeps proteins far from isoelectric point: high net charge, maximum water binding (dry surface, firm), closed structure absorbs light and oxygen poorly (dark, little oxymyoglobin bloom)	—
Water-holding capacity	Very low; drip loss 6–10 percent	Very high; little drip	Moderate; 2–4 percent
Colour	Pale, pinkish-grey	Dark purple-red	Bright red / pink
Microbial shelf life	Normal (low pH)	Short: high pH and available glucose exhausted favour proteolytic spoilage bacteria (Pseudomonas), off-odours early	Normal
Processing consequences	Poor cooked ham yield, poor emulsion stability, poor cured colour, high cooking loss	Unsuitable for fresh retail (rejected by consumers), vacuum-packed meat spoils; useful in cooked and emulsion products because of high WHC, but cured colour is poor	—

Feature	PSE (pale, soft, exudative)	DFD (dark, firm, dry; dark-cutting beef)	Normal
Detection	pH at 45 min (pH45) below 6.0; colour (L* value high); drip loss; conductivity (PQM meter); halothane and DNA test for RYR1 in breeding stock	pHu above 6.0 at 24 h; colour (L* low); glucose oxidase strip for muscle glucose	—

Mitigation strategies

Point of control	PSE	DFD
Genetics	Remove halothane-positive (RYR1) animals; select against stress-susceptible lines	Select calm temperament
Transport	Short, cool, low stocking density; avoid heat; avoid electric goads	Minimise duration; avoid mixing unfamiliar animals; provide water; avoid cold and wet; keep sexes separate
Lairage	Rest 2–4 hours with showers in warm weather to calm and cool; do not over-rest (which depletes glycogen)	Rest with feed where lairage is prolonged, but the cure is to prevent glycogen depletion before arrival; mixing in lairage worsens DFD
Stunning and slaughter	Effective stunning (CO_2 or electrical), rapid bleeding; avoid pre-stunning excitement	Same
Chilling	Rapid chilling of pork carcasses (blast chilling, −20 °C air, or spray chilling) to slow glycolysis before pH falls too far at high temperature	Normal chilling; electrical stimulation does not help because glycogen is absent
Processing	Use PSE meat in emulsion products with phosphate, blend with normal meat	Use DFD meat in cooked, cured, or emulsion products; do not vacuum-pack as fresh

[HPSC EXAM TRAP] PSE = acute stress = rapid pH fall = low ultimate pH; DFD = chronic stress = little glycogen = high ultimate pH. A pair giving PSE a high ultimate pH or DFD a rapid pH fall is wrong. Both have poor cured colour, but only DFD has poor microbial shelf life.

3.4 Meat preservation and processing

Curing chemistry

Principle: Curing is the combined action of salt, nitrite (or nitrate reduced to nitrite) and reductants to produce a product that is microbiologically safer, pink, flavoured and resistant to rancidity.

Ingredient	Level	Function	Mechanism
Sodium chloride	2–3 percent in product; brine 10–20 percent	Water-activity reduction (a_w below 0.97 at 3 percent slows many bacteria); flavour; extracts myofibrillar proteins for binding; synergistic with nitrite	Osmotic dehydration of microbial cells; chloride ion toxicity; increases swelling of myofibrils above pH 5.5 (higher WHC)

Ingredient	Level	Function	Mechanism
Sodium nitrite	Ingoing 100–150 ppm (mg/kg) typical; statutory maxima vary (for example 200 ppm in many codes); never exceed limits because nitrite is toxic (lethal dose about 1 g)	(1) Inhibits Clostridium botulinum spore outgrowth and toxin formation (the anti-botulinal function is the primary safety reason for nitrite in canned cured meats and packaged hams); (2) colour; (3) cured flavour; (4) antioxidant delaying warmed-over flavour	Nitrite → nitrous acid (pH 5.5–6) → nitric oxide (NO) with reductants; NO binds haem iron of myoglobin to form nitrosomyoglobin (red); heating denatures globin, giving stable pink nitrosohaemochrome. NO and nitrous acid react with iron-sulphur enzymes (ferredoxin) of clostridia, blocking spore germination (Perigo effect: heated nitrite is more inhibitory). Nitrite also chelates iron, limiting lipid oxidation
Sodium or potassium nitrate	Dry-cured products (hams, fermented sausages)	Reservoir of nitrite in long cures	Reduced to nitrite by nitrate-reducing bacteria (Staphylococcus carnosus, Micrococcus) over days to weeks
Sodium ascorbate or erythorbate (isoascorbate)	500–550 ppm	Cure accelerator; colour stability; blocks N-nitrosamine formation	Reduces nitrite to NO rapidly and keeps haem iron reduced; scavenges nitrosating species so that secondary amines are not nitrosated during frying (bacon)
Phosphates (sodium tripolyphosphate)	0.3–0.5 percent	Raise pH and ionic strength, chelate calcium and magnesium, dissociate actomyosin	Increase WHC and yield of hams and emulsion sausages
Sugar	1–2 percent	Flavour, counteracts salt harshness, substrate for fermentation	Browning

Curing methods: dry cure (rub, weeks), brine immersion (days), pumping (artery or multi-needle stitch injection with 10–20 percent pickle followed by tumbling or massaging to distribute brine and extract proteins), and direct addition in comminuted products. Nitrosamine concern: nitrite plus secondary amines at high temperature (fried bacon) forms N-nitrosopyrrolidine; controlled by limiting ingoing nitrite, mandatory ascorbate, and avoiding over-cooking.

Smoking

Principle: Smoke is the product of incomplete combustion (pyrolysis) of hardwood (oak, hickory, beech; softwoods with resins are avoided) at 200–400 °C with limited oxygen. The fractions deposited on the meat surface preserve, colour and flavour it.

Smoke constituent	Source in wood	Effect on product
Phenols (guaiacol, 4-methylguaiacol, syringol, eugenol)	Lignin pyrolysis	Antimicrobial (surface), antioxidant (delays rancidity), smoky flavour
Carbonyls (formaldehyde, acetaldehyde, furfural, hydroxyacetaldehyde)	Cellulose and hemicellulose pyrolysis	Maillard-type browning with surface amino groups gives the golden-brown smoke colour; formaldehyde cross-links collagen in casings (skin formation); some antimicrobial effect
Organic acids (acetic, formic,	Hemicellulose	Lower surface pH, inhibit bacteria, aid

Smoke constituent	Source in wood	Effect on product
propionic)		skin formation and peelability of casings
Polycyclic aromatic hydrocarbons (benzo[a]pyrene)	High-temperature pyrolysis above 400 °C, fat dripping into fire	Carcinogenic contaminant; limited by controlled smoke generation at lower temperature, filtering, and by use of liquid smoke (aqueous condensate with PAH removed); EU limit for benzo[a]pyrene in smoked meat 2 µg/kg

Smoking regimes: cold smoking (15–30 °C, days; salmon, dry sausages; no cooking, product remains raw), warm smoking (30–50 °C), hot smoking (60–90 °C with cooking to a core of 70–72 °C; frankfurters, hams). Drying before smoking (surface tacky) is essential for uniform deposition. Smoking is a surface treatment and is combined with curing, drying and refrigeration for safety; smoke alone is not a sterilising process.

3.5 Seafood technology and freshness indices

Principle: Fish muscle differs from mammalian muscle in three ways that make it spoil faster: small glycogen reserves (so the ultimate pH stays high at 6.2–6.8, which is favourable to bacteria), a higher proportion of easily degraded proteins and very active endogenous enzymes adapted to cold-water temperatures (so autolysis continues on ice), and a surface flora of psychrotrophic Gram-negative bacteria (Pseudomonas, Shewanella) that grow at 0 °C. Fish also carry trimethylamine oxide (TMAO) as an osmolyte, which bacteria convert to the fishy-smelling trimethylamine.

Post-harvest autolysis cascade

Fish rigor begins within minutes to a few hours (faster if struggled on the hook or stored warm) and resolves within 1–3 days on ice. Nucleotide degradation runs in parallel:

ATP ⟶ ADP ⟶ AMP ⟶ IMP ⟶ HxR (inosine) ⟶ Hx (hypoxanthine)
(ATPase) (myokinase) (AMP deaminase) (5'-nucleotidase, endogenous) (nucleoside phosphorylase, mainly bacterial)

- ATP, ADP and AMP disappear within about 24 hours.
- IMP (inosine monophosphate) is responsible for the pleasant fresh-fish umami flavour; its accumulation is the peak of eating quality (1–3 days on ice).
- Inosine is flavourless; hypoxanthine is bitter and signals spoilage. The IMP → inosine step is autolytic; the inosine → hypoxanthine step is accelerated by bacteria, so the ratio tracks both autolysis and microbial activity.

Freshness indices

Index	Definition / formula	Interpretation	Notes
K-value	$K\,(\%) = \dfrac{[\text{HxR}] + [\text{Hx}]}{[\text{ATP}] + [\text{ADP}] + [\text{AMP}] + [\text{IMP}] + [\text{HxR}] + [\text{Hx}]} \times 100$	Below 20 percent: very fresh (sashimi grade); 20–60 percent: moderately fresh, acceptable for cooking; above 60 percent: spoiled. Ki-value (simplified: HxR and Hx over IMP + HxR + Hx) is used after 24 hours	Species-specific rates; HPLC or enzyme-strip (Fresh Tester) measurement; best index for the early (autolytic) phase because it rises before TVB-N
Total volatile basic nitrogen (TVB-N)	Sum of ammonia, dimethylamine and trimethylamine nitrogen, mg N per 100 g, by Conway microdiffusion or steam	Fresh: 5–20 mg; rejection: above 30–35 mg N/100 g (EU limits 25, 30 or 35 mg depending on species; many Indian references use 30	Rises late (bacterial deamination of amino acids), so it is a spoilage index rather than a freshness index; high in

Index	Definition / formula	Interpretation	Notes
	distillation	mg)	elasmobranchs (urea) and in frozen fish (DMA from TMAO breakdown by endogenous TMAOase)
Trimethylamine nitrogen (TMA-N)	TMA formed by bacterial reduction of TMAO; measured by picrate method, mg N per 100 g	Fresh: below 1.5 mg; rejection: above 10–15 mg N/100 g	Reaction: TMAO + NADH (or lactate) → TMA + H2O, catalysed by TMAO reductase of Shewanella putrefaciens (the specific spoilage organism of iced marine fish), Photobacterium phosphoreum (vacuum-packed fish), Vibrio and Aeromonas; freshwater fish contain little TMAO so TMA is not a useful index for them; TMAO is an anaerobic electron acceptor, so TMA formation continues even in vacuum packs
Histamine	Decarboxylation of histidine by Morganella morganii, Klebsiella, Hafnia in scombroid fish (tuna, mackerel, sardine) held above 4 °C	Limits of 50 mg/kg (US FDA action level), 100–200 mg/kg (EU, Codex); FSSR specifies limits for fish and fish products	Causes scombroid poisoning; heat-stable, not destroyed by cooking or canning
Peroxide value and TBA number	Lipid oxidation in fatty fish	PV above 10–20 meq O2/kg or TBA above 1–2 mg malonaldehyde/kg indicates rancidity	Omega-3 PUFA are highly susceptible; glazing and vacuum packing protect
Sensory (EU or Torry scheme) and Torrymeter	Eyes, gills, skin, odour, texture; dielectric properties of skin	Grades E, A, B, unfit	Rapid, non-destructive
Microbial	Total viable count at 20–25 °C	Above 10^6–10^7 cfu/g: spoilage	Slow (48 h)

Preservation essentials

Icing at 1:1 fish to flaked ice (0 °C) gives 10–14 days for lean fish; chilled seawater and slurry ice chill faster; superchilling (−1 to −2 °C); freezing at −30 to −40 °C in plate or blast freezers with glazing (ice coat) to prevent dehydration and oxidation, stored at −18 °C or below; modified atmosphere (CO2 40–60 percent) extends chilled life; canning (tuna, sardine) at 115–121 °C; drying, salting and smoking traditional. Freshness is lost most during the first hours after catch, so onboard chilling matters more than any later treatment.

[HPSC EXAM TRAP] K-value is a freshness index that rises from day zero; TVB-N and TMA-N are spoilage indices that rise late. A pair stating that TVB-N is the best indicator of very fresh fish is wrong. TMA comes from TMAO by bacterial reduction, not from protein breakdown, and is irrelevant in freshwater fish. Histamine is not destroyed by cooking.

3.6 Egg processing and quality evaluation

Structure and chemistry

Component	Share of egg weight	Structure and composition	Functional and safety role
Shell	9–12 percent	94 percent calcium carbonate (calcite), 1 percent magnesium carbonate and phosphate, 4 percent organic matrix; 7,000–17,000 pores (funnel-shaped, more at the broad end); outer cuticle (bloom) of mucin-like protein seals pores; two shell membranes (keratin-like) separate at the broad end to form the air cell	Gas exchange (CO_2 out, O_2 in); cuticle and membranes are the first barriers to bacteria; washing removes cuticle (hence oiling after washing); air cell grows as water and CO_2 are lost
Albumen (egg white)	58–60 percent	Four layers: outer thin, thick (gel), inner thin, chalaziferous; about 88 percent water, 10–11 percent protein, 0.5 percent carbohydrate, almost no fat; pH 7.6 at lay	Foaming (meringue), coagulation, antimicrobial defence
Yolk	30–33 percent	Enclosed in vitelline membrane; alternating light and dark layers around the latebra; about 50 percent water, 32–35 percent lipid (triglycerides, phospholipid lecithin, cholesterol), 16 percent protein (lipovitellin, lipovitellenin as low-density lipoprotein, phosvitin, livetins); carotenoid pigments (lutein, zeaxanthin) from feed; pH 6.0	Emulsification (mayonnaise), colour, richness; main site of Salmonella Enteritidis multiplication when the vitelline membrane weakens
Chalazae	—	Twisted cords of thick albumen (ovomucin) anchoring the yolk to the poles	Keep yolk centred; prominent chalazae indicate freshness

Albumen proteins and their functions

Protein	Share of albumen protein	Property
Ovalbumin	54 percent	Phosphoglycoprotein; main structural and foaming protein; denatures on heating (coagulation at 80–84 °C); converts to heat-stable S-ovalbumin on storage
Ovotransferrin (conalbumin)	12–13 percent	Binds iron (and copper, zinc), starving bacteria of iron: bacteriostatic; most heat-sensitive albumen protein (denatures at 57–65 °C), which limits pasteurisation temperature; stabilised by adding aluminium or iron salts in commercial pasteurisation

Protein	Share of albumen protein	Property
Ovomucoid	11 percent	Trypsin inhibitor; heat-stable; major allergen
Ovomucin	1.5–3.5 percent	Sulphated glycoprotein fibres forming the gel of thick albumen; its complex with lysozyme maintains thick-white viscosity
Lysozyme (ovoglobulin G1)	3.4 percent	N-acetylmuramidase that lyses Gram-positive cell walls; antimicrobial; used as a food preservative (cheese)
Ovoglobulins G2, G3	8 percent	Foaming
Avidin	0.05 percent	Binds biotin extremely tightly (antinutrient in raw white; destroyed by cooking); antimicrobial by biotin deprivation
Ovoinhibitor, cystatin, ovoflavoprotein	Minor	Protease inhibitors, riboflavin carrier (yellow-green tint of white)

Quality grading

Candling: the egg is rotated in front of a bright light in a dark room (or on a conveyor with a light bank). It reveals shell cracks and dirt, air-cell depth, yolk shadow visibility and mobility, blood and meat spots, and development. Air-cell depth is measured at the broad end with a gauge.

Grade (USDA, candled)	Air cell depth	Albumen	Yolk	Shell
AA	Not more than 1/8 inch (about 3 mm)	Clear, firm; yolk outline indistinct	Practically free of defects, centred, barely visible	Clean, unbroken, normal shape
A	Not more than 3/16 inch (about 5 mm)	Clear, reasonably firm; outline fairly well defined	Practically free of defects	Clean, unbroken
B	More than 3/16 inch (some texts: more than 8 mm); may be free or bubbly	Weak, watery; yolk outline plainly visible	Enlarged, flattened, slightly mobile; small spots permitted	Slightly stained; abnormal shape permitted

Haugh unit (HU) — Raymond Haugh, 1937 — the standard interior quality measure, relating the height of thick albumen (H, in mm, measured with a tripod micrometer beside the yolk after breaking the egg on a flat glass) to egg weight (W, in grams), so that eggs of different sizes can be compared:

$$HU = 100 \log_{10}\left(H - 1.7\,W^{0.37} + 7.57\right)$$

Interpretation: AA 72 and above; A 60–71; B 31–59; below 30 unfit. A freshly laid egg scores 80–90; HU falls by about 1 unit per day at room temperature and much more slowly at 10 °C. Example: for a 60 g egg with albumen height 7.0 mm, $1.7 \times 60^{0.37} = 1.7 \times 4.55 = 7.73$; $H - 7.73 + 7.57 = 6.84$; $\log_{10} 6.84 = 0.835$; HU = 83.5 (Grade AA).

Yolk index = yolk height / yolk width (both in mm, measured on the broken-out egg). Fresh: 0.38–0.44 (typical 0.40–0.42); falls as the yolk absorbs water from the white and the vitelline membrane weakens; below 0.25 the yolk flattens and breaks easily on separation (stale). **Albumen index** = thick-albumen height / average diameter (fresh 0.09–0.12). **Yolk colour** by Roche (DSM) fan 1–15.

Deterioration chemistry during storage

1. **Carbon dioxide loss through pores:** albumen pH rises from 7.6 at lay to 9.0–9.5 within a few days at room temperature (yolk pH rises from 6.0 to 6.4–6.8). This is the primary trigger of all other changes; it is slowed by cool storage, oiling the shell (mineral oil seals pores), coating, or storage in CO2-enriched atmospheres.
2. **Thinning of thick albumen:** at high pH the ovomucin-lysozyme electrostatic complex dissociates and the ovomucin gel network breaks (β-ovomucin loses carbohydrate), so thick white becomes thin, watery white; Haugh unit falls; the yolk moves freely and the outline becomes visible on candling.
3. **Water migration:** water moves from the white into the yolk by osmosis; the yolk enlarges and flattens; the vitelline membrane stretches and weakens (yolk index falls); the yolk becomes easier to break and more accessible to any bacteria that penetrate the white.
4. **Air-cell enlargement:** water loss (0.5–1 percent weight loss per week at ambient) enlarges the air cell; the egg floats in brine when stale (specific gravity below 1.06).
5. **Chemical flavour changes:** off-flavours from lipid oxidation, absorption of odours through pores, conversion of ovalbumin to S-ovalbumin (poorer foaming and gel strength).
6. **Microbial rots:** bacteria enter via pores when the cuticle is damaged or the egg is wet (thermal contraction draws contaminated water in, so eggs must not be washed in water cooler than the egg). The albumen's defences (lysozyme, ovotransferrin, avidin, high pH) are overcome by certain Gram-negative bacteria, and the yolk, once reached, supports rapid growth.

Rot	Organism	Appearance and odour
Green rot	Pseudomonas fluorescens (and other pseudomonads)	Green fluorescent pigment in the white, visible under UV light at early stage; later sour, fruity odour; most common rot of chilled eggs
Black rot	Proteus vulgaris (also Aeromonas, some Pseudomonas)	Yolk and white turn muddy brown to black; gas (hydrogen sulphide, putrid odour); eggs may burst
Red rot	Serratia marcescens	Red pigment
Colourless (clear) rot	Alcaligenes, Achromobacter, Pseudomonas	Odourless or mild; detected by candling as watery white
Pink rot	Pseudomonas	Pink white and yolk
Mould (pin-spot or whisker)	Penicillium, Cladosporium, Sporotrichum	Coloured spots on shell or just under it in humid storage
Sour egg	Pseudomonas, Alcaligenes	Sour odour, slight turbidity of white

Salmonella Enteritidis contaminates eggs trans-ovarially (inside the yolk or albumen before the shell forms) as well as through the shell; it does not cause visible rot, which is why intact-looking eggs can carry the pathogen and why pasteurisation and cooking are required.

Preservation and processing

- **Cool storage:** 10–13 °C and 70–80 percent relative humidity for market eggs (prevents sweating and mould); 0–1 °C with 85–90 percent RH for long storage up to 6–9 months (with oiling).
- **Shell treatments:** oiling with odourless mineral oil within 24 hours of lay (seals pores, halves HU loss), thermostabilisation (dipping in hot water or oil at 54–60 °C for 10–15 minutes forms a thin coagulated layer of albumen under the shell and kills surface bacteria), coating with waxes or shellac, immersion in lime water or water glass (sodium silicate; traditional).
- **Pasteurisation of liquid egg:** whole egg 60 °C for 3.5 minutes (USDA; 64.4 °C for 2.5 minutes alternative); egg white 56–57 °C for 3.5 minutes (lower because ovotransferrin is heat-labile; stabilised with aluminium sulphate or by adjusting pH to 7.0); yolk 61–63 °C for 3.5 minutes (fat

protects bacteria, so a higher temperature is needed); target is a 9-log reduction of Salmonella; α-amylase test confirms whole-egg pasteurisation.

- **Dried egg:** spray drying after desugaring (glucose removed by glucose oxidase-catalase or by yeast or bacterial fermentation) to prevent Maillard browning and loss of solubility in storage; egg-white powder is pasteurised in the dry state (hot room at 54 °C for 7 days) to destroy Salmonella.
- **Frozen egg:** yolk gels on freezing below −6 °C (lipoprotein aggregation); prevented by adding 10 percent salt or sugar before freezing.
- **Egg products:** liquid, frozen, dried; used in bakery, mayonnaise, noodles.

[HPSC EXAM TRAP] Albumen pH rises on storage (7.6 → 9.5) because carbon dioxide is lost; it does not fall. Haugh unit and yolk index both fall with age. Green rot is Pseudomonas, black rot is Proteus; items swap them. The air-cell thresholds increase from AA to B, and HU thresholds decrease from AA to B.

Part 4: Horticultural Processing

Part 4. Horticultural Processing (Fruits and Vegetables)

4.1 Post-harvest physiology

Principle: A harvested fruit or vegetable is alive. It continues to respire, transpire, ripen and senesce using its own stored reserves, because it has been cut off from the plant's supply of water, sugars and minerals. Post-harvest technology is the management of this living tissue to slow its metabolism without killing it.

Respiration

$$\text{C}*6*|*text{H}*{12}\text{O}*6 + 6*|,|*text{O}*2 \rightarrow 6\,\text{CO}*2 + 6*|,|*text{H}*2\text{O} + \text{energy (about 2,816 kJ, i.e. 673 kcal per mole of glucose)}$$

About 40 percent of the energy is captured as ATP; the remainder is released as **vital heat** (heat of respiration), which must be removed by the cold store refrigeration load. Vital heat is proportional to respiration rate: leafy vegetables, broccoli, asparagus, mushrooms and berries (high respiration, 20–60 mg CO2/kg/h at 5 °C) are highly perishable and generate much heat; apples, citrus, onions, potatoes and nuts (low respiration, 1–10 mg CO2/kg/h) store for months.

- **Respiratory quotient (RQ)** = CO2 evolved / O2 consumed: about 1.0 for carbohydrate substrate, below 1 for lipids (avocado), above 1 when organic acids are oxidised or when fermentation begins (anaerobic conditions, RQ rises sharply with ethanol and acetaldehyde accumulation).
- **Temperature coefficient Q10** = (rate at T + 10 °C) / (rate at T). For most produce Q10 is 2–3 in the 5–25 °C range (and higher, up to 4–7, between 0 and 10 °C). Consequence: each 10 °C reduction halves to thirds the respiration rate and roughly doubles or triples the storage life. Pre-cooling immediately after harvest (hydrocooling, forced-air cooling, vacuum cooling for leafy crops, package icing) removes field heat and gives the largest single gain in shelf life.
- **Chilling injury:** tropical and subtropical produce (banana below 13 °C, mango below 10 °C, tomato below 10–12 °C, cucumber and capsicum below 7 °C) suffers membrane phase change at low but non-freezing temperature, with pitting, uneven ripening, browning and decay. Each commodity therefore has a lowest safe temperature.

Transpiration

Water loss by evaporation from the surface, driven by the vapour pressure deficit between the saturated intercellular air of the tissue and the storage air. Rate is governed by surface-to-volume ratio (leafy vegetables lose fastest), surface structure (cuticle thickness, waxiness, stomata, lenticels, wounds), temperature and relative humidity, and air velocity. A loss of 5–10 percent of fresh weight produces visible wilting, shrivelling and loss of crispness and saleable weight. Control: high RH (85–95 percent; 95–100 percent for leafy crops), low temperature (lowers vapour pressure), reduced air movement once cooled, waxing (carnauba, shellac on citrus, apple, cucumber), film packaging and liners, curing of root crops (healing of wounds).

Climacteric versus non-climacteric fruits

Feature	Climacteric	Non-climacteric
Respiration pattern	After a pre-climacteric minimum, respiration rises sharply to a climacteric peak that coincides with ripening, then declines in senescence	Respiration declines gradually with no peak; ripening changes are slow and continuous

Feature	Climacteric	Non-climacteric
Ethylene	Autocatalytic burst (System 2): ethylene induces its own synthesis; internal ethylene rises 100–1,000 fold; exogenous ethylene advances the climacteric and hastens ripening	Low, non-autocatalytic (System 1) production; exogenous ethylene increases respiration transiently but does not trigger true ripening (it does degreen citrus)
Harvest stage	Can be harvested mature-green and ripened off the plant (storage and transport advantage)	Must be harvested when ripe or nearly ripe because they do not improve after harvest
Starch reserves	Often high; starch converts to sugar during ripening (banana 20 percent starch to 1 percent)	Low; sugars accumulate only on the plant
Examples	Banana, mango, apple, pear, tomato, papaya, guava, sapota (chiku), avocado, kiwifruit, apricot, peach, plum, fig, passion fruit, custard apple, jackfruit, muskmelon (cantaloupe)	Citrus (orange, lemon, lime, grapefruit), grape, strawberry, pineapple, cherry, litchi, pomegranate, cucumber, watermelon, cashew apple, capsicum, olive, most vegetables

[HPSC EXAM TRAP] Ripening and the climacteric peak are not the same as senescence; respiration rises during the climacteric and then falls. Exogenous ethylene makes non-climacteric fruit respire more but does not make it climacteric. Items that place pineapple, grape or citrus in the climacteric list are wrong; items that place avocado or banana in the non-climacteric list are wrong. Watermelon and cucumber are non-climacteric; muskmelon is climacteric.

Ethylene biosynthesis and management

Methionine —(SAM synthetase)→ S-adenosyl methionine (SAM) —(ACC synthase; rate-limiting, induced by ripening, wounding, stress)→ 1-aminocyclopropane-1-carboxylic acid (ACC) —(ACC oxidase; needs O_2)→ Ethylene (C_2H_4)

Ethylene is a gaseous hormone active at 0.1–1 µL/L (ppm). It binds copper-containing ETR receptors; binding relieves repression and ripening genes (polygalacturonase, pectin methylesterase, cellulase, chlorophyllase, ACC synthase, carotenoid and volatile synthesis enzymes) are expressed. Ethylene is also produced by wounds, decay, engines (exhaust) and by all ripening fruit; hence mixed storage of apples with leafy vegetables causes yellowing, and of bananas with other fruit causes premature ripening.

Strategy	Mechanism	Practical use
1-Methylcyclopropene (1-MCP, SmartFresh)	A cyclic olefin that binds the ethylene receptor irreversibly (about 10 times higher affinity than ethylene) so that ethylene present cannot act; cells must synthesise new receptors before responsiveness returns (days to weeks)	Applied as gas at 0.5–1 µL/L for 12–24 hours in a sealed room at 20 °C before storage; apples (keeps crispness 3–6 months), bananas, avocados, cut flowers; GRAS, no residue; effect depends on maturity at treatment
Potassium permanganate ($KMnO_4$) scrubbers	Strong oxidant on an inert porous carrier (alumina, zeolite, vermiculite; Purafil) oxidises ethylene to CO_2 and water; the purple carrier turns brown when exhausted	Sachets in cartons of bananas and mangoes; filter beds in cold rooms; cheap, but must not contact food directly ($KMnO_4$ is toxic)
Ventilation and low temperature	Dilutes ethylene; cold slows both ethylene synthesis and action	Basic cold-chain practice; one air change per hour in storage
Controlled and modified atmosphere	Low O_2 limits ACC oxidase (needs oxygen); high CO_2 competitively inhibits ethylene action	See below
Catalytic, UV or ozone scrubbers	Oxidise ethylene	Commercial cold stores

Strategy	Mechanism	Practical use
Genetic (antisense ACC synthase, ACC oxidase)	Reduce ethylene production	Flavr Savr type tomatoes (historical)
Controlled ripening with ethylene or ethephon	Ethylene gas at 10–100 µL/L for 24–48 h at 18–20 °C and 90 percent RH in ripening rooms; ethephon (2-chloroethylphosphonic acid) releases ethylene at pH above 4 in spray or dip; calcium carbide (acetylene, industrial grade contaminated with arsenic and phosphorus) is prohibited under FSSR	Banana, mango, citrus degreening (2–5 µL/L)

4.2 Controlled atmosphere (CA) and modified atmosphere packaging (MAP)

Principle: Lowering oxygen and raising carbon dioxide around the produce slows respiration (the oxidase enzymes are substrate-limited), suppresses ethylene synthesis and action, slows chlorophyll degradation, softening and decay, and reduces chilling injury of some crops. The atmosphere is a supplement to, never a substitute for, refrigeration.

Feature	Controlled atmosphere (CA)	Modified atmosphere packaging (MAP)
Scale	Gas-tight storage rooms or sea containers (tonnes)	Consumer or bulk packs (grams to kilograms)
Control	Atmosphere is continuously measured and actively regulated: nitrogen flushing or generators lower O2, CO2 scrubbers (lime, activated carbon, molecular sieves) remove excess CO2, ethylene scrubbers; dynamic CA adjusts O2 to the fermentation threshold using chlorophyll fluorescence or RQ	Atmosphere is set once (active MAP: gas flush; passive MAP: created by the produce's own respiration and the film's permeability) and then drifts to an equilibrium determined by the balance of respiration and film gas transmission; no further control
Typical gases	O2 2–5 percent (as low as 1–1.5 percent for apples in ultra-low oxygen CA), CO2 3–10 percent (1–3 percent for apples, up to 10–20 percent for strawberries short term), balance N2, at 0–3 °C and 90–95 percent RH	O2 3–10 percent, CO2 5–15 percent in equilibrium; films: LDPE, LLDPE, PP, microperforated films chosen so that O2 permeability matches respiration rate; CO2 permeability of polymer films is 3–5 times that of O2
Commodities	Apples (6–12 months), pears, kiwifruit, cabbage, bananas in transit, avocados, pomegranate	Fresh-cut salads, broccoli, mushrooms, strawberries, bananas in transit, fresh-cut fruit; shelf life gains of 50–100 percent at chill temperature
Risks	Low O2 below the commodity's tolerance (about 1 percent) → anaerobic fermentation, ethanol and acetaldehyde, off-flavour; high CO2 → brown heart (apple), internal browning (pear), surface pitting; both worsened by poor temperature control	Temperature abuse raises respiration faster than film permeability, so O2 collapses and CO2 rises → fermentation, off-odour, and the possibility of anaerobic pathogens (Clostridium botulinum in low-acid vegetables); condensation; leakage
Cost	High capital, suited to long storage	Low cost per pack, suited to retail distribution

Hypobaric (low-pressure) storage reduces O2 partial pressure and removes ethylene continuously but is costly. Edible coatings and shrink-wrapping are individual modified atmospheres.

4.3 Thermal canning (appertisation)

Principle: Canning destroys spoilage and pathogenic microorganisms by heat inside a hermetically sealed container so that re-contamination cannot occur. The severity of the heat process is set by the most heat-resistant pathogen capable of growing in the product, and that depends on pH: Clostridium botulinum cannot grow at pH 4.6 or below, so acid foods need only 100 °C, while low-acid foods (pH above 4.6, aw above 0.85) must receive a botulinum cook at 121 °C.

Unit-operation sequence

1. **Selection and preparation:** sound, uniform maturity (peas at tenderometer reading, mango firm-ripe); washing (chlorinated water 25–50 ppm); sorting, grading; peeling (hand, steam, lye at 1–2 percent NaOH at 95–100 °C for 1–3 minutes for peaches and tomatoes, abrasive for root crops); cutting, coring, pitting.
2. **Blanching:** short heat treatment in hot water (85–100 °C, 1–5 minutes) or steam, followed by cooling (except before hot filling). Purposes: inactivate enzymes that would otherwise cause off-flavour and colour changes during the come-up time and in any under-processed cold spot (lipoxygenase, polyphenol oxidase, pectin methylesterase, catalase, peroxidase); expel intercellular gases (reduces can corrosion, improves vacuum and prevents floating); wilt and soften tissue for filling; fix colour (chlorophyll) and remove raw flavour; reduce microbial load; aid skin removal. **Peroxidase test:** peroxidase is the most heat-resistant of the quality-related plant enzymes and is easily measured (guaiacol plus hydrogen peroxide turns brown within 1–3 minutes if peroxidase survives), so its absence proves that all other relevant enzymes are also inactivated; catalase is used as an alternative, less stringent indicator. Over-blanching leaches sugars, vitamins (ascorbic acid losses 10–40 percent) and minerals and softens texture; calcium chloride (0.1–0.5 percent) is added to blanch water for firmness of tomatoes and beans.
3. **Filling:** hand or machine; **syrup** (20–40 °Brix for fruits; grades light, medium, heavy) or **brine** (1–2 percent salt for vegetables); fill weight and headspace (6–9 mm) controlled; drained weight is a standard.
4. **Exhausting:** heating the filled, open can in a hot-water or steam exhaust box so that the centre reaches 80–85 °C (some references 82–90 °C for 5–10 minutes), or hot filling at above 80 °C, or mechanical vacuum closing. Purposes: expel air from the headspace and tissue so that, after sealing and cooling, a partial vacuum (250–400 mm Hg) forms inside the can; the vacuum prevents bulging during processing and at altitude, minimises oxidation (discolouration, vitamin C loss), reduces internal corrosion and strain on seams, and lets the ends stay concave as a visible indicator of soundness. Vacuum is a consequence of exhausting, not of the retort.
5. **Sealing (double seaming):** the can lid (end) is mechanically interlocked with the body flange in two operations: the first-operation roll curls the end hook under the body hook; the second-operation roll flattens and tightens the hooks together with the sealing compound (rubber-based) in the end curl, giving a hermetic seam of five thicknesses of metal (three from the end, two from the body). Seam quality is checked by body hook, cover hook, overlap, tightness and seam thickness; glass jars use lug or vacuum caps.
6. **Thermal processing (retorting):**

Food class	pH	Target organism and process	Temperature and typical time
High-acid and acid foods (fruits, fruit juices, tomato products below 4.6 after acidification, pickles)	4.6 or below (high-acid below 3.7)	Yeasts, moulds, lactic and butyric bacteria, Bacillus coagulans (flat-sour in tomato juice); botulinum spores cannot outgrow	100 °C (boiling water bath or atmospheric steam) for 10–45 minutes; hot-fill-hold at 90 °C for juices
Low-acid foods (peas, beans, corn, carrots, mushrooms, meat, fish, milk)	Above 4.6	Clostridium botulinum spores: minimum botulinum cook of F0 = 3 minutes (12-D reduction, D121 of 0.21	115–121 °C (10–15 psi steam) for 20–90 minutes depending on can size, product consistency

Food class	pH	Target organism and process	Temperature and typical time
		min); commercial sterility usually needs F0 of 5–15 minutes to also control thermophilic spoilers (Geobacillus stearothermophilus D121 about 4 min, Clostridium thermosaccharolyticum, Desulfotomaculum nigrificans)	(conduction versus convection heating) and initial temperature; process established by heat-penetration tests (cold spot thermocouple) and calculated by the General or Ball formula methods

Retort types: vertical and horizontal batch still retorts (steam, with air venting), rotary (agitating, convection heating, shorter time), continuous hydrostatic and reel-and-spiral cookers, water-immersion and steam-air overpressure retorts for pouches and glass.

1. **Cooling:** immediately after the process, cans are cooled under water (with overpressure for large cans to prevent seam strain) to an average of 38–40 °C. Reasons: stop cooking (texture, colour, nutrient losses, stack burning); prevent germination and growth of surviving thermophilic spores (which grow at 45–65 °C: flat-sour and swells in warm warehouses); but not below 38 °C, so that the residual heat dries the can surface and prevents rusting. Cooling water must be chlorinated (2–5 ppm residual) because the seam compound is momentarily soft and microscopic leakage can draw water in (post-process leaker spoilage, the most common cause of canned-food contamination).
2. **Labelling, incubation and storage:** representative cans incubated at 37 °C for 10 days (and 55 °C for thermophiles) to check for swells; storage below 35 °C, dry.

Can enamels (lacquers)

Enamel	Chemistry	Problem prevented	Products
Plain (unlacquered) tinplate	Tin coating over steel; tin dissolves slowly (a sacrificial, non-toxic reaction) and keeps the product light coloured	Tin acts as a reducing agent, preserving colour of light fruits	Peaches, pears, pineapple, citrus segments, apricots (light fruits that discolour in lacquered cans)
C-enamel (sulphur-resistant)	Oleoresinous or epoxy enamel pigmented with zinc oxide (and sometimes zinc carbonate)	Sulphur compounds released from proteins during heating react with iron and tin to form black iron sulphide and purple tin sulphide stains; zinc oxide traps sulphide as white zinc sulphide	Corn, peas, beans, meat, fish, mushrooms, asparagus, cabbage
R-enamel (acid-resistant, fruit enamel)	Oleoresinous enamel, no zinc oxide (zinc would dissolve in acid)	Prevents acid from dissolving tin and iron (which bleaches anthocyanins and causes hydrogen swells) and protects colour of red fruits	Berries, cherries, plums, red grapes, beetroot and other strongly coloured acid products
Epoxy-phenolic, vinyl, organosol, aluminium-pigmented	Modern synthetic lacquers with high chemical resistance	Universal; some two-coat systems	Tomato products, beverages, aerosol foods; lead-free, BPA-free specifications increasing
Tin-free steel (TFS), aluminium	Chromium-coated steel; aluminium with lacquer	Corrosion of acidic products	Beverages, fish

Canned-food spoilage diagnostics

Type	Cause	Can appearance	Product
Flat sour	Bacillus coagulans (acid foods), Geobacillus stearothermophilus (low-acid); under-processing or slow cooling, warm storage	Flat (no swell)	Sour, cloudy, no gas
Thermophilic anaerobe (TA) spoilage	Clostridium thermosaccharolyticum	Hard swell, may burst	Sour, cheesy odour, gas (CO_2, H_2)
Sulphide stinker	Desulfotomaculum (Clostridium) nigrificans	Flat (H_2S dissolves)	Blackened, rotten-egg odour
Putrefactive anaerobe	Clostridium sporogenes (PA 3679, the non-toxic surrogate for botulinum in process design), C. botulinum	Swell	Putrid; botulinum may show no change
Hydrogen swell	Acid attack on tin and iron, long storage, high nitrate, poor lacquer	Swell, pinholes	Metallic flavour, high tin
Leaker spoilage	Post-process contamination through seam	Any; mixed flora	Variable
Springer, flipper, soft swell, hard swell	Degrees of end bulging	Progressive	Diagnostic of gas pressure

[HPSC EXAM TRAP] Exhausting produces the vacuum; retorting produces sterility. The pH dividing line is 4.6: items that quote 4.5 or 5.0, or that assign a 121 °C process to fruit in syrup, are wrong. C-enamel is for sulphur-rich foods (corn, peas, fish) and R-enamel is for acid, coloured fruits; light fruits are canned in plain tin. Cooling is to 38–40 °C, not to room temperature, so that the can dries.

4.4 Freezing

Principle: Freezing preserves by converting water to ice (removing it as a solvent and lowering water activity) and by low temperature slowing all reactions. It does not sterilise: most bacteria survive, some are killed, and enzymes remain partly active, which is why vegetables are blanched before freezing and why frozen fruits are packed with sugar or ascorbic acid.

- **Ice crystal size and quality:** slow freezing (hours; home freezer) forms large extracellular crystals that rupture cell walls, draw water out of cells, and cause heavy drip loss, mushiness and nutrient leaching on thawing. Fast freezing (minutes; passing the critical zone of −1 to −5 °C, where 70–80 percent of water freezes, in under 30 minutes) forms many small intracellular crystals that preserve structure. Recrystallisation (growth of large crystals at the expense of small ones) occurs during temperature fluctuation in storage, so a constant −18 °C or lower is needed.
- **Individual quick freezing (IQF):** each piece (pea, corn kernel, berry, shrimp, diced carrot, mango cube) is frozen separately, so the pack is free-flowing and the consumer can take out a portion. Achieved in fluidised-bed freezers (cold air at −30 to −40 °C blown upward through a perforated belt at a velocity that suspends the pieces; freezing time 3–10 minutes), belt tunnel freezers, impingement freezers, or cryogenic freezers (liquid nitrogen at −196 °C, or solid CO_2 at −78 °C; freezing in seconds to a few minutes; least drip loss, highest cost; used for shrimp, berries, cut fruit). IQF requires blanched vegetables and firm fruit; glazing (spraying water to form an ice coat) protects IQF shrimp and fish from freezer burn.

- **Other freezers:** air-blast (batch rooms and tunnels, −30 to −40 °C, 3–5 m/s), plate (contact, for rectangular blocks of spinach or fish), immersion (brine or glycol for packaged products), scraped-surface (for juices).
- **Freezer burn:** sublimation of ice from the surface of unprotected product in dry freezer air, leaving a dry, white, spongy, toughened surface; prevented by vapour-tight packaging, glazing and full packs.
- **Thawing** should be slow at refrigeration temperature or in the cooking step (vegetables cooked from frozen) to limit drip and microbial growth.

4.5 Dehydration

Principle: Drying removes water until water activity falls below about 0.6 (microbial growth stops below 0.6; enzymes and non-enzymatic browning are minimal around 0.2–0.4). The engineering problem is to remove water fast enough to prevent spoilage and slow enough to avoid damaging the product.

Drying kinetics

- **Constant-rate period:** the surface is wet; water evaporates at the wet-bulb temperature as fast as the air can carry it away; rate depends on air temperature, humidity and velocity, not on the product.
- **Critical moisture content:** the point at which the surface is no longer saturated.
- **Falling-rate period(s):** water must diffuse from the interior to the surface; the rate depends on product thickness, structure and temperature; the product temperature rises toward the air temperature, so heat damage is most likely late in drying. Most of the drying time is spent here.
- **Equilibrium moisture content:** the moisture at which the product is in equilibrium with the drying air; the lowest achievable at that temperature and humidity (reason for finishing in a lower-humidity bin).

Case hardening

If the initial drying is too fast (very hot, very dry air on a sugar- or starch-rich product such as fruit slices or potato), the surface dries and shrinks into a hard, dense, impermeable skin while the interior is still wet. Sugars and dissolved solids carried to the surface by capillary flow crystallise or form a glassy film; proteins and pectins denature and gel at the surface. The skin seals in moisture: internal water can no longer diffuse out, the drying rate collapses, the centre remains moist and later spoils (mould) or the product cracks and splits as the interior finally shrinks. Prevention: moderate initial air temperature and higher initial humidity (or a two-stage schedule: humid-warm then dry-hot), thinner and uniform slices, adequate pretreatment (sulphuring, blanching) and dipping in alkaline emulsions (dipping grapes in ethyl oleate-potassium carbonate removes the waxy cuticle to speed drying uniformly).

Methods and pretreatments

Method	Conditions	Products	Remarks
Sun drying and solar drying	Ambient, 10–30 percent moisture end point difficult	Raisins, apricots, dates, chillies, fish	Cheap; contamination, slow, weather-dependent; solar cabinets improve hygiene
Cabinet (tray) and tunnel dryers	50–70 °C, 1–3 m/s air; counter-current or concurrent flow	Vegetables (onion, garlic, cabbage), fruit slices	Most common commercial method for horticultural products
Fluidised-bed and belt dryers	60–90 °C	Peas, diced vegetables, granules	Uniform
Drum (roller) dryer	Thin film on steam-heated drum at 120–150 °C, seconds	Mashed potato flakes, tomato flakes, baby food	Some heat damage, cheap
Spray dryer	150–200 °C inlet	Fruit and vegetable powders	Stickiness of sugar-rich

Method	Conditions	Products	Remarks
		(tomato, mango) with carriers (maltodextrin)	juices requires carriers and cool walls
Foam-mat drying	Juice foamed with stabilisers, dried in thin layer	Juice powders	Short time, good quality
Osmotic dehydration	Immersion in 50–70 °Brix syrup, removes 30–50 percent water without heat	Pre-step for fruit candying, freeze-dried or air-dried fruit	Improves colour and flavour retention
Freeze drying (lyophilisation)	Product frozen, then ice sublimed under vacuum (below 611 Pa, the triple-point pressure) with gentle heat; primary and secondary drying to 2 percent moisture	Strawberries, mushrooms, herbs, instant soups, coffee	Best quality and rehydration, porous structure, most expensive; product must be packed in inert gas because it is highly oxidisable
Sulphuring and sulphiting	Burning sulphur (1–3 kg per tonne, 2–4 hours in a sulphur box) or dipping in 0.2–0.5 percent potassium metabisulphite to give 1,000–2,000 ppm SO2 in apricots, 500–1,000 ppm in raisins and apple rings	Light-coloured fruits	Inhibits polyphenol oxidase and non-enzymatic browning, preserves ascorbic acid and carotene, insect protection; residual SO2 limits under FSSR; destroys thiamine
Blanching before drying	As for canning	Vegetables	Inactivates enzymes; dried vegetables not blanched develop hay-like flavour

Quality indices of dried products

- **Dehydration ratio** = fresh weight / dried weight (for example 8–12 for leafy vegetables, 4–6 for onion).
- **Rehydration ratio** = weight of rehydrated product / weight of dried product (good products 4–8); **coefficient of rehydration** compares with the original fresh weight; low values indicate case hardening, over-drying or heat damage.
- Colour (browning index), sulphur dioxide content, moisture (vegetables 4–8 percent, fruits 15–25 percent), bulk density, and absence of mould.

[HPSC EXAM TRAP] Case hardening is caused by rapid initial drying at high temperature and low humidity, and it slows drying; it is not caused by slow drying and it is not the same as over-drying or freezer burn. Freeze drying is sublimation (solid to vapour) below the triple point, never melting. Blanching is required before freezing and drying of vegetables, but fruits are usually not blanched (they are sulphured or sugared instead).

4.6 Comparative table: preservation technologies for horticultural produce

Technology	Principle	Typical conditions	Microbial effect	Main quality risk
Refrigerated storage	Lower temperature slows respiration (Q10) and microbes	0–13 °C depending on chilling sensitivity, 85–95 percent RH	Slows, does not kill	Chilling injury, moisture loss
CA / MAP	Low O2 and high CO2 slow respiration and ethylene	2–5 percent O2, 3–10 percent CO2 (CA); film-equilibrium	Slows aerobes; risk of anaerobes if abused	Fermentation, CO2 injury

Technology	Principle	Typical conditions	Microbial effect	Main quality risk
		(MAP)		
Canning	Heat in hermetic container	100 °C (acid) or 121 °C (low-acid)	Commercial sterility	Over-cooking, enamel and seam failures
Freezing	Water to ice, low temperature	−18 °C or lower; IQF at −30 to −40 °C	Stops growth; survival	Ice-crystal damage, freezer burn, enzyme activity without blanching
Dehydration	Water activity below 0.6	50–70 °C air; sulphuring	Stops growth; survival of spores	Case hardening, browning, vitamin loss
Chemical and minimal processing (fresh-cut, waxing, coatings, SO2, sorbate in juices)	Barrier or inhibitor	Varies	Partial inhibition	Residues, limited shelf life

Revision Bank & Model Answers

Stage 1 Revision Bank and Stage 2 Model Answers

A. Stage 1 count-based pair bank (exactly four items)

Protocol: score each pair $c_i \in \{0, 1\}$, compute $N = c_1 + c_2 + c_3$, map N to None / Only 1 / Only 2 / All 3.

Item 1 — Thermal processing of milk

Consider the following process–condition pairs:

1. HTST pasteurisation — 72 °C for 15 seconds
2. LTLT pasteurisation — 63 °C for 15 seconds
3. UHT processing — 135–150 °C for 1–4 seconds

How many pairs are correctly matched?

Verification:

- Pair 1: HTST is 72 °C (71.7 °C in Indian practice) for 15 seconds in a plate heat exchanger with holding tube. Correct. $c_1 = 1$.
- Pair 2: LTLT (holder) is 63 °C for 30 minutes, not 15 seconds; the unit has been changed from minutes to seconds. Incorrect. $c_2 = 0$.
- Pair 3: UHT is 135–150 °C for 1–4 seconds followed by aseptic packaging. Correct. $c_3 = 1$.
- $N = c_1 + c_2 + c_3 = 1 + 0 + 1 = 2$.

Answer: Only 2. The trap is the unit substitution in pair 2; the temperature was left correct to lure a careless reader.

Item 2 — Wheat roller milling

Consider the following pairs:

1. Break rolls — corrugated surface with a speed differential of about 2.5 : 1
2. Reduction rolls — corrugated surface with a speed differential of about 1.25 : 1
3. Maida (straight-grade refined flour) — extraction rate of 70–72 percent

How many pairs are correctly matched?

Verification:

- Pair 1: Break rolls are fluted (corrugated) and run at the high differential of about 2.5 : 1 to shear the kernel open. Correct. $c_1 = 1$.
- Pair 2: Reduction rolls have a smooth surface; the differential of 1.25 : 1 is correct but the surface description is wrong, and one wrong attribute makes the pair wrong. Incorrect. $c_2 = 0$.
- Pair 3: Maida extraction is 70–72 percent; atta is 95–100 percent; patent flour 40–60 percent. Correct. $c_3 = 1$.
- $N = 1 + 0 + 1 = 2$.

Answer: Only 2. Note the half-truth design of pair 2: a correct number attached to an incorrect attribute.

Item 3 — Post-mortem muscle biochemistry

Consider the following pairs:

1. PSE meat — results from chronic pre-slaughter stress that exhausts muscle glycogen
2. DFD meat — ultimate pH above 6.2 and a dark, dry cut surface
3. Resolution of rigor mortis — actin–myosin cross-bridges are released because ATP is resynthesised after 24 hours

How many pairs are correctly matched?

Verification:

- Pair 1: Chronic stress with glycogen exhaustion is the aetiology of DFD, not PSE. PSE results from acute stress (and the halothane RYR1 genotype) causing a rapid pH fall while the carcass is still warm. Incorrect. $c_1 = 0$.
- Pair 2: DFD is defined by high ultimate pH (above 6.0, commonly 6.2), high water-holding capacity (dry surface), closed structure (dark). Correct. $c_2 = 1$.
- Pair 3: ATP is never resynthesised post-mortem; rigor resolves by proteolysis of the cytoskeletal framework by calpains and cathepsins, not by release of the actomyosin bond. Incorrect. $c_3 = 0$.
- $N = 0 + 1 + 0 = 1$.

Answer: Only 1.

Item 4 — Fish and egg quality indices

Consider the following pairs:

1. K-value below 20 percent — very fresh fish suitable for raw consumption
2. Trimethylamine nitrogen — formed by reduction of trimethylamine oxide by Shewanella putrefaciens
3. Haugh unit of 72 or above — Grade AA egg

How many pairs are correctly matched?

Verification:

- Pair 1: K-value = (inosine + hypoxanthine)/(total ATP-breakdown products) × 100; below 20 percent is sashimi grade. Correct. $c_1 = 1$.
- Pair 2: TMAO is reduced to TMA by TMAO reductase of Shewanella putrefaciens (and Photobacterium phosphoreum in vacuum packs); the fishy odour is TMA. Correct. $c_2 = 1$.
- Pair 3: USDA interior grades by Haugh unit: AA 72 and above, A 60–71, B 31–59. Correct. $c_3 = 1$.
- $N = 1 + 1 + 1 = 3$.

Answer: All 3. Candidates conditioned to expect a trap in every item lose marks here; score each pair independently.

Quick-recall absolute-qualifier traps (for the same objective paper)

Statement	Verdict	Reason
Homogenisation always reduces lipolysis	False	It increases susceptibility to lipolysis by replacing the native membrane with casein
ALP-negative milk is always sterile	False	ALP negative proves pasteurisation; turbidity negative proves sterilisation
Parboiling always lowers the nutritive value of rice	False	Thiamine and mineral retention rise
All climacteric fruits can be harvested mature-green	Broadly true	They ripen off the plant with ethylene
Exogenous ethylene never affects non-climacteric fruit	False	It degreens citrus and raises respiration transiently without true ripening
C. botulinum never grows below pH 4.6	Accepted basis of the acid/low-acid division	Hence 100 °C versus 121 °C processing
All egg rots are caused by Pseudomonas	False	Black rot is Proteus; red rot Serratia
Freezing kills all microorganisms	False	Growth stops; most survive

B. Stage 2 model answer — 10 marks

Question

Detail the step-by-step industrial sequence of wheat roller milling. Contrast the mechanical actions and extraction products of break rolls versus reduction rolls.

Model answer (about 11 minutes; 450–550 words plus one table and one flow sheet)

Introduction and principle. Wheat roller milling is a gradual-reduction process whose objective is to separate the starchy endosperm (about 83 percent of the kernel) from the bran (14–15 percent) and germ (2–3 percent) as completely as possible, and then to reduce the endosperm to flour of the required particle size with minimum bran contamination (measured as ash) and controlled starch damage. The process relies on the fact that properly conditioned bran is tough and tears into large flakes, while the endosperm is friable and crumbles.

Step-wise industrial sequence.

1. *Receiving and cleaning (screen room):* magnetic separators remove iron; milling separators (sieves plus aspiration) remove straw, dust and chaff; destoners remove stones by specific gravity; disc or cylinder separators remove other seeds by length; scourers remove surface dirt and beeswing.
2. *Conditioning and tempering:* water is added to raise moisture to 14–15 percent for soft wheat and 15.5–16.5 percent for hard wheat, and the grain rests in tempering bins for 12–24 hours (shorter for soft wheat). Water toughens the bran so that it does not shatter into specks and mellows the endosperm so that it separates cleanly from the bran and grinds easily.
3. *Break system (B1–B5):* the tempered wheat passes through four to six pairs of corrugated (fluted) rolls running at a speed differential of about 2.5 : 1. The fast roll shears the kernel open against the slow roll and scrapes the endosperm off the bran in progressively finer stages; flutes increase from about 10–12 per inch on B1 to 26–28 per inch on B5, and roll gaps narrow at each stage. After each break the stock is sifted.
4. *Sifting (plansifters):* stacked sieves separate each break product into overtails (bran with adhering endosperm, sent to the next break), coarse semolina and fine middlings (sent to purifiers), and a small amount of break flour.
5. *Purification:* the purifier is an oscillating sieve deck with an upward air current. Light bran chips of the same size as semolina particles are lifted off by aspiration while the denser endosperm particles pass through the graded sieves. This step is what makes low-ash white flour possible.
6. *Sizing and reduction system (C1–C12):* purified semolina and middlings pass through eight to twelve pairs of smooth rolls at a differential of about 1.25 : 1. The particles are crushed, not sheared, into flour; the germ, being oily and plastic, is flattened into flakes and sifted out; residual fine bran (shorts) is removed as tailings.
7. *Flour collection, treatment and packing:* flour streams are blended into patent, straight-grade (maida, 70–72 percent extraction) and clear grades; treated with permitted improvers (ascorbic acid, enzymes, fortification with iron, folic acid and vitamin B12); and packed. Whole-wheat atta (95–100 percent extraction) is produced by recombining bran and germ with the flour or by stone (chakki) milling.

Flow sheet (in words): Wheat → cleaning → conditioning (14–16.5 percent moisture, 12–24 h) → Break 1 → sifter → Break 2 … Break 5 → bran out; semolina and middlings → purifiers → sizing rolls → reduction rolls C1 … C12 with sifters → flour streams → blending and treatment → maida, patent flour, suji, bran, shorts, germ.

Contrast of break and reduction rolls.

Attribute	Break rolls	Reduction rolls
Roll surface	Corrugated, flutes 10–28 per inch	Smooth (matt)
Speed differential	About 2.5 : 1	About 1.25 : 1
Mechanical action	Shearing and cutting; opens the kernel and scrapes endosperm from bran	Crushing and compression; grinds clean endosperm to flour
Number of passages	4–6	8–12 (plus sizings)
Feed	Whole wheat, then bran overtails	Purified semolina and middlings
Principal products	Bran flakes, semolina (suji/rava), middlings, little flour	Flour (patent, straight, clear), fine bran/shorts, germ
Quality objective	Maximum large bran flakes, minimum bran powder	Fine flour with controlled starch damage (6–9 percent) and minimum ash

Extraction products. Maida 70–72 percent; patent flour 40–60 percent; semolina recovered from break stock; bran 14–15 percent; shorts and germ 2–4 percent; whole atta 95–100 percent. Higher extraction raises ash, fibre, fat and B-vitamins but darkens the flour and shortens shelf life because of germ oil.

Conclusion. Roller milling succeeds by separating before reducing: the break system and purifiers isolate clean endosperm, and the reduction system then produces flour of the ash and granulation demanded by the end use, with the extraction rate as the key economic and nutritional control variable.

Suggested marks distribution: sequence and parameters 5; break versus reduction contrast 3; extraction products and conclusion 2.

C. Stage 2 model answer — 15 marks

Question

Examine the post-mortem biochemical changes occurring in muscle tissue from slaughter to the resolution of rigor mortis. Explain the aetiology of PSE and DFD meat defects and evaluate strategies to mitigate them.

Model answer (about 16–17 minutes; 750–900 words with two tables)

Introduction and principle. Conversion of muscle to meat is the running-down of living muscle metabolism after the blood supply stops. Three linked processes determine meat quality: anaerobic glycolysis and pH decline, ATP depletion leading to rigor mortis, and proteolytic resolution of rigor during conditioning. Abnormalities in the rate or extent of the first process produce the PSE and DFD defects.

1. Immediate post-slaughter phase (0–30 minutes). Exsanguination removes oxygen delivery. Muscle keeps consuming ATP to maintain ion gradients and relaxation. Creatine phosphate regenerates ATP briefly; thereafter the only ATP source is anaerobic glycolysis of muscle glycogen (0.5–1.3 percent).

2. Post-mortem glycolysis and pH decline (30 minutes to 24 hours). Glycogen is converted via glucose-1-phosphate and the Embden–Meyerhof pathway to pyruvate and then to lactic acid, yielding two or three ATP per glucose unit. Without circulation, lactic acid accumulates and pH falls from about 7.0–7.2 to an ultimate pH of 5.4–5.8 within 6–8 hours in pork, 2–4 hours in poultry breast and 18–24 hours in beef. Glycolysis stops when glycogen is exhausted or when low pH inactivates phosphofructokinase. Consequences of the pH fall: proteins approach their isoelectric point (about 5.1–5.3), net charge and water-holding capacity (WHC) fall, the muscle structure opens and scatters more light (paler colour), oxymyoglobin bloom develops on cutting, flavour precursors form, and the meat becomes microbiologically more stable than living tissue.

3. Rigor mortis (onset 1–12 hours; full rigor 3–24 hours). When ATP falls below about 1 µmol/g, the sarcoplasmic reticulum can no longer pump calcium; calcium leaks into the sarcoplasm, myosin heads

bind actin, and because detachment requires ATP the cross-bridges lock permanently as actomyosin. The muscle loses extensibility and becomes stiff; WHC reaches its minimum. The degree of sarcomere shortening depends on temperature at ATP exhaustion: cold shortening occurs if pre-rigor muscle is chilled below 10 °C (calcium pump failure in the cold, contraction up to 40 percent, severe toughness), and heat shortening at above 35 °C; thaw rigor follows thawing of pre-rigor frozen muscle.

4. Resolution of rigor: conditioning or ageing (1–21 days). The actomyosin bonds are not reversed. Instead, endogenous proteases weaken the structural scaffold around them. Calpains (μ- and m-calpain, calcium-activated neutral proteases most active early post-mortem at pH 6–7 and regulated by calpastatin) degrade titin, nebulin, desmin and troponin-T, loosening the Z-line and the myofibril–membrane linkages. Cathepsins (lysosomal acid proteases B, D, H and L) are released as pH falls and contribute during longer ageing. The result is progressive tenderisation, partial recovery of WHC and development of aged flavour over 7–21 days at 1–4 °C for beef, 3–5 days for pork and about one day for poultry. Electrical stimulation of carcasses accelerates glycolysis so that rigor is complete before chilling, preventing cold shortening and speeding ageing.

5. Aetiology of PSE and DFD.

Feature	PSE (pale, soft, exudative)	DFD (dark, firm, dry)
Species	Mainly pork; turkey and chicken breast	Mainly beef and lamb (dark-cutting beef); pork
Stress type	Acute, immediately before slaughter (fighting, rough handling, goads, heat); genetic predisposition through the halothane-sensitive RYR1 ryanodine-receptor mutation (porcine stress syndrome)	Chronic, over 12–48 hours (long transport, mixing strange animals, fasting, cold, exhaustion)
Glycogen at death	Normal or high	Depleted before slaughter
pH course	Very rapid fall: pH 5.8–6.0 or lower within 45 minutes while the carcass is still above 35 °C	Little or no fall; ultimate pH remains above 6.0 (6.2 commonly cited)
Mechanism	Low pH at high temperature denatures sarcoplasmic and myofibrillar proteins; denatured sarcoplasmic proteins precipitate onto the myofibrils, scatter light (pale) and lose water binding (soft, exudative; drip loss 6–10 percent)	High pH keeps proteins far from the isoelectric point: high net charge binds water tightly (dry surface, firm), the closed structure absorbs light and limits oxygen penetration so little bright oxymyoglobin forms (dark)
Consequences	Poor cooked yield, emulsion stability and cured colour	Short shelf life (high pH and no residual glucose favour proteolytic Pseudomonas spoilage), consumer rejection of fresh cuts, poor cured colour, unsuitable for vacuum packing
Detection	pH at 45 minutes, colour (L*), drip loss, conductivity, halothane or DNA test	pH at 24 hours, colour, muscle glucose strip

6. Evaluation of mitigation strategies.

Control point	Measure	Evaluation
Genetics	Eliminate RYR1 halothane carriers from pig breeding; select calm temperament in cattle	Highly effective and permanent for PSE; DNA testing is cheap; does not address environmental stress
On-farm handling and transport	Low stocking density, trained handlers, no electric goads, short cool journeys, avoid mixing unfamiliar animals, provide water	Reduces both defects; the most cost-effective measure for DFD; requires enforcement across the supply chain

Control point	Measure	Evaluation
Lairage	2–4 hours rest with showers in warm weather	Calms acutely excited pigs (reduces PSE); prolonged lairage with fighting worsens DFD; feeding sugar before slaughter is sometimes used to restore glycogen
Stunning and bleeding	Effective CO2 or electrical stunning, rapid bleeding, no pre-stun excitement	Reduces acute stress; welfare and quality aligned
Chilling	Rapid or spray chilling of pork carcasses to slow glycolysis before pH falls at high temperature	Reduces PSE severity; must be balanced against cold shortening in beef and lamb, where electrical stimulation and conditioning at 15 °C until pH falls below 6 are preferred
Processing use	Blend PSE meat with normal meat in phosphate-containing emulsions; direct DFD meat to cooked, cured and emulsion products where its high WHC is an asset	Salvage, not prevention; cured colour remains inferior in both

Conclusion. Post-mortem quality is governed by how fast and how far the pH falls relative to carcass temperature and by subsequent proteolysis. PSE is the product of too fast a fall while the muscle is warm; DFD is the product of too little fall because glycogen was consumed by chronic stress. Because both defects originate before the knife, the most effective mitigation is genetic selection combined with humane, low-stress handling, supported by correct chilling and electrical stimulation at the plant.

Suggested marks distribution: glycolysis and rigor onset 4; resolution and proteolysis 3; PSE and DFD aetiology 5; mitigation evaluation 2; conclusion 1.

D. Rapid-fire numeric recall sheet

Fact	Value
Milk specific gravity	1.028–1.032
Milk titratable acidity	0.12–0.16 percent lactic acid
Milk freezing point	